Votes for the Victims? Organizational Ecologies and the Territorial Limits of Electoral Insulation in Colombia's Peace Constituencies

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Abstract

Why do identical institutional rules produce different outcomes across space? Colombia's peace constituencies were designed to insulate victim representation from traditional parties through formal exclusion, yet 2022 elections across 167 municipalities diverged sharply: some achieved autonomous mobilization while others tracked concurrent congressional elections with suspicious precision. This paper argues that electoral insulation is not an institutional property but a territorial achievement, contingent on local configurations of state capacity, organizational density, and political competition. Using a three-stage diagnostic strategy—design-aligned residuals to identify anomalous comovement, Random Forest classification with SHAP explanations to reveal predictive interactions, and Latent Class Analysis to generate interpretable typologies—we identify six influence ecologies where insulation fails, partially holds, or succeeds. Organizational variables dominate the predictive hierarchy, accounting for over half of total explanatory weight. Insulation holds where community institutions provide

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autonomous gatekeeping but fails where capable local states have been instrumentalized by machine networks operating through legal administrative channels. The findings introduce stratified autonomy to describe how uniform institutions produce patterned rather than random permeability across municipalities. The diagnostic framework offers a portable method for evaluating reserved seats and special districts wherever territorial heterogeneity precludes causal identification, with direct implications for indigenous constituencies in Peru, special interest groups in Uganda, and scheduled caste reservations in India. Understanding why identical institutions function differently across space is not merely an empirical exercise but a democratic imperative for post-conflict states.

1 Introduction

In 2022, Colombia held elections for sixteen Circunscripciones Transitorias Especiales de Paz (CITREP)—special congressional districts designed to give victims of armed conflict direct representation while excluding traditional political parties. The institutional design was uniform: parties were legally barred, candidacies were restricted to registered victim organizations, and districts targeted municipalities with the highest conflict exposure. Yet across the 167 municipalities where elections were held, outcomes diverged sharply. In some territories, victim organizations mobilized autonomously and secured representation consistent with the reform's normative intent. In others, CITREP vote shares tracked concurrent congressional elections with suspicious precision, suggesting that traditional political networks had penetrated spaces explicitly designed to exclude them. The puzzle is not whether insulation failed in the aggregate, but why identical institutional rules produced radically different outcomes across municipalities governed by the same formal design.

This paper argues that the variation cannot be explained by rule violations, individual misconduct, or generalized elite capture. Instead, it reflects a design-territory interaction failure that produces stratified autonomy—a condition where uniform institutions deliver protection unequally across space, patterned by organizational capacity rather than need. Electoral insulation depends not only on formal exclusion rules but on the territorial environments into which those rules are inserted. Where local conditions—state capacity, organizational density, conflict legacies—supported autonomous mobilization, insulation proved viable. Where they did not, traditional political influence reasserted itself through adaptation rather than violation. Territory is not a neutral container for institutional experimentation; it is a structuring force that amplifies or nullifies formal protections depending on configurations of municipal characteristics that legal rules cannot erase.

We conceptualize these territorial environments as "influence ecologies": recurring configurations of state presence, organizational infrastructure, and political competition that jointly condition the feasibility of autonomous representation. These ecologies operate as systems rather than additive effects. High state capacity protects insulation when paired with cohesive victim movements but enables co-optation when facing

fragmented organizations and powerful extractive economies. Organizational density shields communities from partisan penetration in environments with weak party networks but proves insufficient against well-resourced machines operating through legal administrative channels. The implication is that influence in post-conflict electoral institutions arises from configurations that must be diagnosed empirically rather than deduced from theory in advance.

Rather than advancing causal estimates, the paper adopts a diagnostic strategy. We operationalize the insulation claim embedded in the CITREP design: under effective insulation, electoral outcomes in peace constituencies should be orthogonal to concurrent congressional elections once municipal fundamentals are accounted for. Deviations signal breaches of autonomy. Using a design-aligned baseline regression, we identify anomalous comovement between House and CITREP vote shares and use this deviation to label municipalities where influence is likely present. A Random Forest classifier then learns how municipal characteristics interact to predict influence, allowing for nonlinearities that linear models cannot capture. Shapley Additive Explanations reveal which features systematically shape the probability of influence across municipalities. Finally, Latent Class Analysis translates high-dimensional predictive patterns into six interpretable influence ecologies, providing a typology of territorial environments where insulation fails, partially holds, or succeeds.

The findings overturn conventional institutional design wisdom. Organizational variables—coalition size, victim movement composition, community council density—dominate the predictive hierarchy, accounting for over fifty percent of the model's total explanatory weight. Municipal structural characteristics—poverty, rurality, educational infrastructure—refine rather than replace this organizational logic, marking environments where asymmetries prove most consequential. Conflict exposure variables contribute minimally. The six ecologies that emerge—*Expanding Community Councils*, *Quiet Peripheries*, *Depoliticized Semi-Centers*, *Mobilized Frontiers*, *Institutionalized High-Conflict Enclaves*, and *Deprived Peripheries*—cut across departmental boundaries and produce profound within-district heterogeneity, exposing an aggregation bias in district-level assessments that masks capture.

The paper makes three contributions. Substantively, it advances a territorial theory of institutional performance in post-conflict settings, showing that the success of inclusionary designs depends less on formal exclusion rules than on their alignment with uneven political geographies. Theoretically, it reframes insulation as a conditional property of local political ecologies rather than a rule-based outcome, introducing the concept of stratified autonomy to describe how uniform institutions produce patterned rather than random permeability. Methodologically, it demonstrates how influence can be diagnosed and typologized without causal overclaiming by combining design-aligned benchmarks with interpretable machine learning, offering a portable framework for evaluating reserved seats, special districts, and transitional arrangements in fragmented states.

The argument travels beyond Colombia. Reserved seats and special districts are increasingly deployed in societies marked by deep territorial inequality—indigenous

constituencies in Peru, special interest groups in Uganda, scheduled caste reservations in India, ethnic quotas across sub-Saharan Africa. When these designs ignore subnational variation, they produce not universal inclusion but stratified autonomy: some communities protected by endogenous organizational capacity, others captured by machines operating through legal channels, still others abandoned in representational voids. Understanding why identical institutions function differently across space is not merely an empirical exercise but a democratic imperative. Without diagnostic attention to where autonomy is feasible and where it requires deliberate construction, even carefully crafted reforms risk reproducing the territorial inequalities they were designed to transcend.

The paper proceeds as follows. Section 2 describes the CITREP institutional design and the territorial constraints it confronted. Section 3 develops the conceptual framework linking organizational ecologies to insulation failure. Section 4 presents our theoretical expectations. Section 5 explains the three-stage diagnostic strategy and data. Section 6 reports results from the baseline regression, Random Forest classification, and Latent Class Analysis. Section 7 interprets the findings and derives implications for post-conflict institutional design. Section 8 concludes by situating the argument within broader debates about democratic inclusion in fragmented states.

2 Institutional Design and Territorial Constraints of the CITREP

The CITREP were created under Colombia's 2016 Final Peace Agreement as a mechanism of political reparation for victims of the armed conflict. Their normative objective was explicit, namely to provide direct congressional representation to populations historically excluded from national politics while insulating that representation from traditional parties and professional politicians. The design rested on the premise that meaningful democratic inclusion required not only access to electoral competition, but protection from the organizational and clientelistic advantages of established political actors.

To achieve this goal, the reform introduced sixteen single-member districts covering 167 municipalities selected on the basis of conflict intensity, rurality, and weak state presence. These municipalities largely overlap with the Territorially Focused Development Programs (PDET), identified in the Peace Agreement as priority zones for institutional reconstruction. Candidate eligibility was restricted to victims' organizations, peasant organizations, social movements, women's organizations, and ethnic organizations. Traditional parties were formally excluded, and each list was limited to two candidates competing for a single seat by majority vote.

While normatively coherent, this design embedded the CITREP within a highly constrained territorial and electoral environment. First, the peace districts were not aligned with departmental boundaries. Most CITREP combine municipalities from two or more departments, reflecting conflict geographies rather than administrative

divisions. Second, the elections for CITREP seats were held concurrently with elections for the House of Representatives. Voters in CITREP municipalities therefore received ballots for Senate, House, and peace seats on the same day, and conventional congressional campaigns unfolded simultaneously in the same territories.

This combination of territorial overlap and electoral concurrency introduced a structural tension into the insulation strategy. Traditional parties and candidates mobilizing for House elections already possessed organizational infrastructure, local brokers, and voter networks active at the municipal level. The presence of CITREP elections within the same spatial and temporal context created conditions under which these existing networks could be projected into the peace constituencies without formal participation or rule violation.

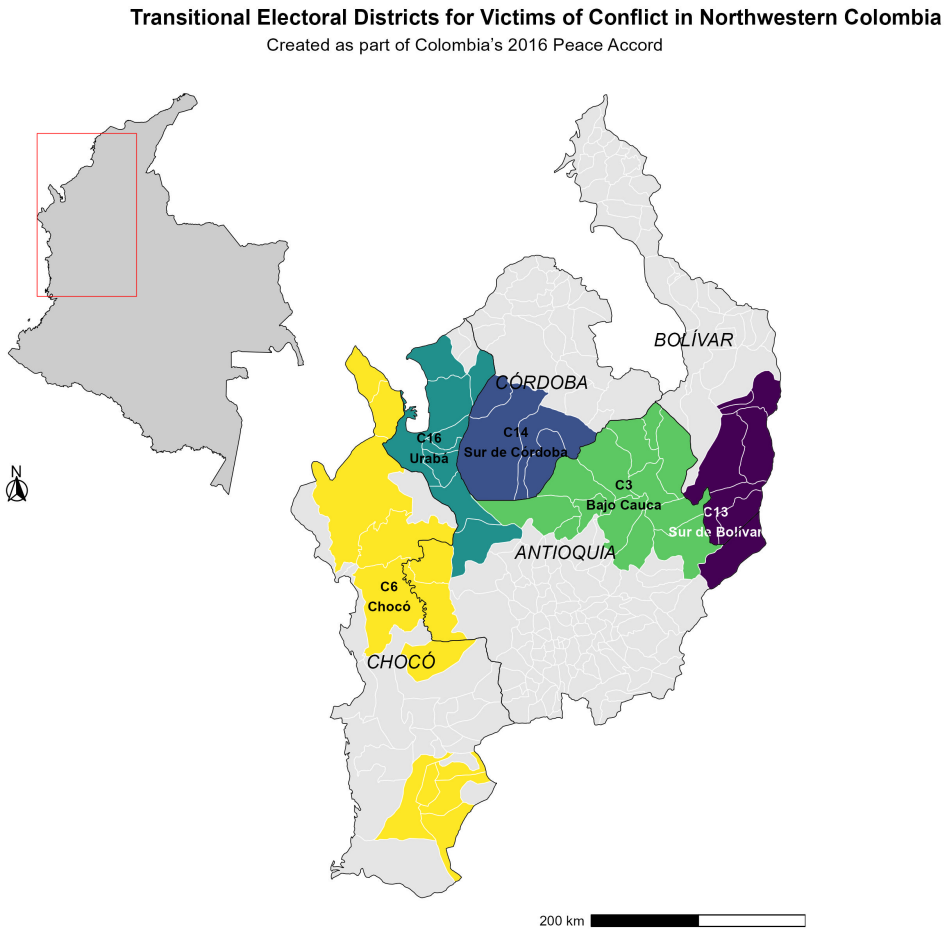

Figure 1: *The map shows how several peace districts overlap with or border the departments of Antioquia, Chocó, Córdoba, and Bolívar. Because CITREP districts are smaller and electorally less demanding, these overlaps created incentives for professional politicians to project existing mobilization networks into adjacent peace constituencies.*

Figure 1 illustrates this dynamic for northwestern Colombia. Several peace districts overlap with, or directly border, the departments of Antioquia, Córdoba, Chocó, and Bolívar—regions with strong and well—established congressional party machines. In a department such as Antioquia, where securing a House seat typically requires close to 100,000 votes, a CITREP seat could be won with fewer than 10,000. This asymmetry in electoral thresholds, combined with spatial adjacency, created a clear incentive structure in which actors already mobilizing votes for departmental races could, at relatively low marginal cost, channel resources, endorsements, or organizational support toward peace lists located within or near their areas of influence.

These conditions did not guarantee influence, nor did they imply systematic rule-breaking. Rather, they made influence plausible and unevenly distributed across space. The feasibility of insulating CITREP elections depended not only on formal exclusion rules, but on the territorial contexts into which those rules were inserted. In municipalities with limited organizational capacity among victim and social groups, weak administrative presence, and dense pre-existing political networks, insulation faced greater practical constraints. In others, where local organizations possessed experience, coordination, and autonomy, the same institutional design could function as intended.

This section establishes the institutional and territorial conditions that shape the problem analyzed in the remainder of the paper. It shows why the question of insulation cannot be addressed solely at the level of rules or intentions, and why variation across municipalities is theoretically central. The next section builds on this context by introducing the concept of influence ecologies, which provides an analytical framework for understanding how these territorial conditions combine to make insulated electoral institutions more or less permeable to traditional political influence.

3 Conceptual Framework

The creation of the CITREP represents a sophisticated attempt at institutional engineering designed to provide "political reparation" to populations historically excluded by armed conflict. The normative core of this design rests on the principle of institutional insulation: the creation of a direct representative channel for victims that bypasses the traditional partisan mediation often characterized by a chronic distance between elites and popular needs (Leal Roncancio 2022). However, the success of such inclusionary designs is not merely a function of formal rules or legal bans on party participation. Rather, insulation must be understood as a territorial property that is fundamentally conditioned by preexisting subnational heterogeneity (Otero-Bahamon 2019). This framework posits that identical institutional rules produce divergent outcomes when inserted into a fragmented landscape of uneven territorial governance, where the promise of autonomy encounters the gravitational pull of entrenched local political bargains and structural power asymmetries.

To understand why institutional insulation fails unevenly, we must shift the analytical lens away from the national-level bias that has long dominated comparative politics

(Eaton 2020; Suárez-Cao, Batlle, and Wills-Otero 2017). A subnational comparative approach is essential because centralized electoral reforms produce highly differential and non-uniform effects across local units (Albarracín and Milanese 2021). In the Colombian context, municipal political trajectories diverge significantly over time, oscillating between stability, opening, and closure depending on the interaction between national designs and local configurations (Pino Uribe 2020). By adopting the municipality as the primary unit of observation, we can move beyond the abstract intent of the CITREP reform to examine how the "design-territory interaction" determines the permeability of institutional firewalls. This requires a framework that accounts for the multi-actor horizontal interactions and asymmetric causalities that define territorial governance in post-conflict settings (Chilito 2018; Suárez-Cao, Batlle, and Wills-Otero 2017).

At the foundation of this territorialized framework sits the role of state capacity, conceptualized not merely as an administrative variable but as the essential infrastructure of democratic viability. A robust state presence is a necessary precondition for democracy, capable of maintaining the order required to protect citizen rights and guarantee autonomous political competition (Fortin 2012). Drawing on the observed J-shaped relationship between administrative capacity and democratization, we expect that interventions aimed at expanding inclusion, such as the CITREP, may encounter their most significant challenges in areas where state capacity is weakest (Bäck and Hadenius 2008). In environments of administrative scarcity, the state's routine visibility declines, and the institutional firewalls meant to protect marginalized groups become more permeable to the influence of established political networks that possess the resources to substitute for missing state functions.

Beyond the infrastructure of the state, the permeability of institutional insulation is shaped by the material reality of local political economies. State weakness and conflict are often mutually reinforcing phenomena within a framework where subnational heterogeneity is explained by differentiated extractive and rentier economic structures (Fergusson 2019). In many post-conflict peripheries, rentier economies—specifically those dependent on non-regulated mining or illicit crops—sustain hegemonic power structures characterized by low levels of political competition (Batlle, Hoyos, and Wills-Otero 2020; Peñaranda Currie, Otero-Bahamon, and Uribe 2021). In these contexts, powerful rural elites and the narcobourgeoisie frequently function as land brokers for large-scale extraction, using coercive dispossession to accumulate wealth and consolidate political influence (Dietz 2023). These structural fragilities are associated with higher permeability of new representative channels, as the materiality of illicit economies actively shapes state formation and the local political machine's capacity to achieve capture.

This structural vulnerability is associated with a process of institutional capture that is better understood as strategic adaptation rather than mere rule violation. Organized crime and entrenched political actors exploit the administrative thinness of local jurisdictions to initiate a scalar capture, establishing what has been termed the co-opted reconfiguration of the state (Galeano Escucha 2018). Within this logic, crim-

inal and partisan actors do not simply bypass institutions; they co-opt governance arrangements—ranging from participatory budgets to specialized electoral districts—by leveraging institutional vulnerability to gain economic benefits and social recognition (Abello-Colak and Guarneros-Meza 2014). Consequently, municipal trajectories diverge significantly: while some areas may experience contingent political openness, others remain trapped in stable hegemonies where the local political machine maximizes the effects of coercion and patronage to maintain its grip on power (Batlle, Hoyos, and Wills-Otero 2020).

The contemporary exercise of influence is further conditioned by the durable legacies of armed governance that characterize Colombia's unequal political geography. During the armed conflict, non-state actors, including insurgents and paramilitaries, often governed alongside communities, acting as substitute providers or brokers for authority in areas of limited state presence (De Bruin et al. 2025; Steele and Weintraub 2022). These historical experiences produce "symbiotic" local orders where the boundaries between state, community, and armed actors are blurred (Peñaranda Currie, Otero-Bahamon, and Uribe 2021). Such governance legacies fundamentally alter civilian expectations and evaluations of legitimacy, which remain based on the procedural coherence and service provision of those who ruled the territory (De Bruin et al. 2025). Moreover, patterns of forced displacement have created segregated enclaves with highly differentiated organizational densities and political legacies (Steele 2019). These historical layers ensure that any new institutional design, no matter how well-insulated in its formal architecture, must navigate a landscape where the imprint of previous armed orders continues to structure political participation and social coordination.

Beyond structural constraints, the persistence of traditional political influence in insulated districts is driven by the strategic resilience of local political machines. Far from retreating in the face of legal firewalls, professional politicians use embedded brokers and established organizational legacies to navigate formal restrictions and territorial boundaries (Otero-Bahamon, Uribe, and Peñaranda-Currie 2022). This adaptive capacity is symptomatic of the chronic distance between national elites and popular needs, where local intermediaries fill the representational gap through clientelistic networks (Leal Roncancio 2022). In this context, the logic of "influence" is not merely a violation of rules but a redeployment of machinery that leverages pre-existing territorial control. Within the ecology of influence, these brokers function as the human infrastructure that bridges the gap between the formal exclusion of traditional parties and the practical necessity of political mobilization in resource-poor environments.

Conversely, the feasibility of political autonomy depends on the presence of counter-ecologies characterized by high organizational density. Autonomous representation is most plausible where social movements and community institutions provide both internal accountability mechanisms and external protection against co-optation (Ley, Mattiace, and Trejo 2019). When social movements achieve coherence rather than fragmentation, they develop the necessary political frames to resist the strategic appropriation of their platforms by external actors (García-Montoya, Güiza-Gómez, and Chang 2025). These systems of regional autonomy serve as a crucial deterrent against

capture; robust social infrastructures allow victim-based organizations to act as independent gatekeepers who can mobilize voter support without relying on the logistical or financial machinery of traditional partisan brokers. Consequently, the "density" of the local organizational environment serves as a primary shielding mechanism that determines the effectiveness of institutional firewalls.

Finally, the specific vulnerability of identity-based or reserved districts highlights a critical balance between group consciousness and institutional performance. While high levels of Racial Group Consciousness or victim-centered identity can initially drive preferences for specialized representation, this preference remains inherently fragile (Villamizar-Chaparro and Echeverri-Pineda 2025). Accountability failures—specifically perceived poor performance or a lack of trust in representative behavior—can lead voters to revert to standard territorial districts or traditional partisan options as a pragmatic fallback. This fragility suggests that identity-based insulation is not a static property of the institutional design but is instead a contingent outcome mediated by the representative's perceived efficacy. In environments of acute deprivation, the failure of identity-based districts to deliver material improvements renders them particularly susceptible to the gravitational pull of professional political networks that offer immediate, if clientelistic, solutions to local needs.

3.1 Configuring Influence: A Municipal-Level Framework

The preceding analysis suggests that political influence in post-conflict settings cannot be understood as the result of isolated variables acting in a linear fashion. Rather, it emerges from a configurational logic where the effects of institutional design are filtered through the complex interaction of state capacity, historical legacies, and social organization. In line with the Subnational Comparative Method's emphasis on asymmetric causalities and non-uniform effects (Albarracín and Milanese 2021; Suárez-Cao, Batlle, and Wills-Otero 2017), this framework posits that the permeability of institutional firewalls is a product of specific territorial alignments. The presence of illicit rents, for instance, is associated with higher rates of co-optation primarily when combined with administrative thinness and fragmented community networks (Batlle, Hoyos, and Wills-Otero 2020; Fergusson 2019). By moving beyond additive models toward a configurational approach, we can better capture the "design—territory interaction" that determines whether new representational channels serve as instruments of democratic renewal or as extensions of entrenched partisan control (Chilito 2018).

This configurational synthesis allows for the formal articulation of the *ecology of influence* as the core unit of analysis. We define an ecology of influence as a recurring municipal-level configuration of territorial characteristics where the formal architecture of institutional insulation—such as the legal bans and eligibility restrictions of the CITREP—is rendered permeable to traditional political actors. This concept shifts the focus from the moral failures of individual actors to the systemic properties of local political orders. An ecology is not merely a geographic container but a relational environment where state visibility, the material base of the local economy, and the density of

community-led intermediation intersect at the municipal level (Otero-Bahamon, Uribe, and Peñaranda-Currie 2022; Peñaranda Currie, Otero-Bahamon, and Uribe 2021). In this view, the "success" or "failure" of an institutional reform is not a global outcome but a localized property of the municipal ecology into which it is embedded, where the institutional design is either reinforced by local social infrastructures or overwhelmed by resilient clientelistic machines.

Justifying the use of ecologies as the primary unit of analysis is essential for a rigorous evaluation of peacebuilding and post-conflict institutional engineering. Standard national-level assessments often flatten the profound heterogeneity of transitional contexts, obscuring the "divergent trajectories" that municipalities experience during the implementation of peace agreements (Pino Uribe 2020). By adopting the ecology of influence as the primary lens, we can diagnose why identical rules for political reparation produce endpoint-sensitive results across territories shaped by "symbiotic" and "hybrid" orders, where governance involves multi-actor horizontal interactions and the state is but one of several competing sources of authority (Chilito 2018; Peñaranda Currie, Otero-Bahamon, and Uribe 2021). Evaluating peace institutions through their constituent ecologies thus provides a granular assessment of the state's capacity to protect the autonomy of marginalized populations amid persistent legacies of conflict and repression (Osorio, Schubiger, and Weintraub 2018). Accordingly, the municipality constitutes the operative analytical scale at which influence ecologies are identified and compared.

This configurational perspective underscores that influence is not a property of isolated variables but of territorial alignments. By embracing this complexity and leveraging interpretable machine-learning tools, the analysis does not seek causal point estimates but rather a systematic characterization of the municipal environments in which electoral insulation is more or less likely to hold. In doing so, it provides a disciplined bridge from institutional design to empirical diagnosis, grounding the study in both formal architecture and territorial reality.

4 Theoretical Expectations

The institutional design of the CITREP rests on a strong insulation claim: the creation of a direct representative channel that bypasses established partisan mediation. As a mechanism of political reparation, the design explicitly excludes traditional parties and restricts eligibility to social and victim organizations to safeguard the autonomy of marginalized communities. Consequently, the baseline theoretical expectation is that the electoral performance of CITREP lists should be statistically orthogonal to the outcomes of concurrent House of Representatives elections within the same territories. If the institutional firewalls are effective, municipal variation in traditional partisan strength should contain no systematic information regarding the electoral success of peace-seat candidates, thereby fulfilling the normative goal of representation without partisan mediation. We operationalize deviations from this baseline as anomalous

comovement between CITREP and House vote shares at the municipal level—the behavioral manifestation of influence. The expectations that follow identify the municipal environments where such anomalous comovement is more likely to occur.

The feasibility of maintaining institutional insulation is fundamentally conditioned by the administrative and infrastructural capacity of the local state. State capacity serves as the essential infrastructure of democratic autonomy; where it is robust, the state provides public goods and maintains order, reducing the representational vacuum often filled by informal brokers (Fortin 2012). Drawing on the observed non-linear relationship between state efficacy and democratization (Bäck and Hadenius 2008), we expect that in environments characterized by "administrative thinness", the formal rules of the CITREP will be more permeable. In such contexts, traditional political networks are more likely to substitute for missing state functions, increasing the probability of anomalous comovement between CITREP and House outcomes.

Beyond state-led infrastructure, the degree of political autonomy achieved in CITREP elections is expected to depend on the internal density and horizontal coordination of the competing social and victim movements. Cohesive organizations with prior experience in advocacy and established internal accountability mechanisms are expected to act as "independent gatekeepers," effectively shielding their candidates from external co-optation (Ley, Mattiace, and Trejo 2019). In contrast, where social movements are fragmented or resource-poor, they are more likely to seek "alliances of convenience" with professional political actors to secure logistical and financial support (García-Montoya, Güiza-Gómez, and Chang 2025). Therefore, we expect that higher organizational density is associated with lower probability of influence, whereas fragmentation is associated with greater strategic appropriation of the CITREP platform by entrenched political machines through informal territorial and organizational control.

The vulnerability of the CITREP design is further conditioned by the material reality of local political economies, specifically the presence of extractive and rentier structures. In many post-conflict peripheries, economic configurations dependent on non-regulated mining or illicit crops like coca sustain local hegemonic power structures characterized by low levels of political competition (Batlle, Hoyos, and Wills-Otero 2020). We expect that in these contexts, traditional political machines will exert greater influence over new representative channels to secure territorial control and manage local rents (Fergusson 2019). This mechanism posits that where powerful rural elites function as land brokers for large-scale extraction (Dietz 2023), the institutional firewalls of the CITREP will be more permeable, as partisan actors leverage structural fragilities to co-opt the representational space for material and political accumulation.

Beyond structural economic constraints, influence is expected to be a function of the territorial reach and organizational legacies of professional party networks. In environments marked by a "chronic distance" between national elites and popular needs (Ramírez and Cobos 2018), embedded brokers are expected to fill the representational void by repurposing their established mobilization machinery to support CITREP candidates. This expectation focuses on the human infrastructure of political machines;

rather than a direct violation of formal rules, influence is projected through the adaptive capacity of brokers who bridge the gap between formal exclusion and the practical requirements of electoral mobilization (Otero-Bahamon, Uribe, and Peñaranda-Currie 2022). Consequently, we expect that the probability of partisan penetration will be higher in municipalities where these informal networks maintain durable territorial control, allowing professional politicians to navigate the formal restrictions of the CITREP design through established brokerage systems.

Finally, the framework anticipates that the effect of state capacity on institutional insulation is contingent on the broader organizational and political context, leading to a "working state" paradox. High levels of administrative capacity do not guarantee autonomy if they are embedded in "hybrid orders" or "co-opted reconfigurations of the state" where efficient service delivery and machine politics function as complements rather than substitutes (Galeano Escucha 2018). This allows for theoretical ambivalence: sophisticated co-optation may occur even in "working" local states where professional networks utilize their organizational infrastructure to project influence (Abello-Colak and Guarneros-Meza 2014). Thus, the final expectation is that insulation is most likely to fail in environments where multi-actor horizontal interactions (Chilito 2018) allow traditional actors to capture the state's formal administrative efficacy for partisan ends, bypassing the intended institutional firewalls.

These expectations reflect a configurational logic rather than an additive model. No single dimension is sufficient to produce influence; rather, the permeability of institutional firewalls emerges from specific combinations of municipal characteristics. State capacity may protect insulation when combined with robust organizational density but prove insufficient when facing powerful extractive economies and fragmented social movements. Similarly, organizational fragmentation may not lead to co-optation in municipalities with weak party networks, while even cohesive movements may face challenges in environments of extreme administrative scarcity. This configurational perspective acknowledges that some dimensions may offset others and that the relationship between municipal characteristics and insulation failure involves nonlinearities and threshold effects that cannot be specified through parametric assumptions alone. The empirical strategy that follows is designed to identify these complex territorial alignments without imposing restrictive functional forms or presuming uniform causal effects across contexts.

Section 5.1 presents the data sources and the quantitative empirical strategy used to evaluate and potentially falsify these hypotheses. By translating these dimensional and mechanistic propositions into a structured diagnostic pipeline, the analysis moves from the conceptual logic of influence ecologies to the empirical detection of anomalous comovement and the characterization of the territorial environments in which it persists.

5 Data and Empirical Strategy

This section describes the data and the empirical strategy used to diagnose and characterize traditional political influence in CITREP elections. The analysis is diagnostic rather than causal. We proceed in three stages. First, we operationalize the insulation claim embedded in the CITREP design by estimating a design-aligned benchmark and using the resulting residuals to label municipalities in which CITREP outcomes display anomalous comovement with concurrent House elections. Second, we use these labels as the outcome in a supervised learning model to identify the municipal features that systematically distinguish influenced from non-influenced cases, emphasizing interpretability rather than predictive performance. Third, we translate the high-dimensional patterns recovered by the classifier into a small number of interpretable influence ecologies through latent class analysis. Together, these steps provide a disciplined bridge from institutional design to empirically grounded typologies of the territorial conditions under which electoral insulation is more or less likely to hold.

5.1 Data

The analysis combines electoral returns from the 2022 legislative elections with municipal-level administrative and socioeconomic data. The unit of analysis is the municipality. The empirical strategy links electoral outcomes in the peace constituencies to concurrent House of Representatives elections and to baseline municipal characteristics common to both contests.

Electoral data for the Circunscripciones Transitorias Especiales de Paz (CITREP) include vote shares obtained by each registered peace list at the municipal level. These data cover all sixteen peace constituencies and the 167 municipalities in which CITREP elections were held. For each municipality, we observe the distribution of votes across CITREP lists competing for the single seat assigned to the corresponding district.

House of Representatives data include municipal-level vote shares for all party lists competing in the ordinary congressional election held on the same day. These data allow us to observe electoral performance by traditional parties and movements in the same territories where CITREP elections took place, under identical voting conditions and turnout environments.

To construct the empirical objects used in the diagnostic baseline, electoral data are organized at the level of municipality-by-pair observations. Within each municipality, we consider all possible pairings between CITREP lists and House lists competing in the concurrent election. Each observation therefore links the vote share of a given CITREP list to the vote share of a given House list in the same municipality. This pairing strategy allows us to assess whether CITREP outcomes exhibit anomalous comovement with any segment of the conventional congressional competition, rather than presuming a priori which actors might be relevant. As a result, the number of observations exceeds the number of municipalities, reflecting the full set of candidate–list pairings evaluated within each locality. Although the diagnostic baseline is estimated

at the municipality–pair level, influence is ultimately classified and analyzed at the municipality level.

Municipal covariates are drawn from official administrative sources and capture baseline structural characteristics that plausibly shape both electoral contests. These include measures of population size, urbanization, poverty, fiscal capacity, state presence, and conflict exposure. All covariates are measured prior to the 2022 elections or are time-invariant, and are included to account for shared municipal fundamentals rather than to model electoral behavior.

All electoral variables are constructed as vote shares at the municipal level. When used in regression-based diagnostics, vote shares are transformed following standard practice to reduce skewness. The resulting dataset provides a unified municipal-level framework linking CITREP outcomes, House electoral performance, and structural characteristics across all peace constituencies.

5.2 Operationalizing Electoral Insulation: Diagnostic Baseline

This subsection operationalizes the insulation claim embedded in the institutional design of the CITREP. The objective is to construct a diagnostic baseline that allows us to identify municipalities in which CITREP electoral outcomes display anomalous comovement with concurrent House of Representatives elections. This step is explicitly diagnostic rather than inferential. It does not seek to estimate causal effects or to attribute influence to specific actors or mechanisms, but to label cases in which the insulation condition implied by the design plausibly fails.

The logic of insulation implies electoral independence between peace-seat contests and ordinary congressional races once shared municipal fundamentals are taken into account. Formally, let y^C_{im} denote the vote share obtained by CITREP list m in municipality i, y^H_{ij} the vote share obtained by House list j in the same municipality, and X_i a vector of baseline municipal controls common to both elections. Under effective insulation, the following conditional independence restriction should hold:

$$\mathbb{E}\big[y^C_{im} \mid y^H_{ij}, X_i\big] = \mathbb{E}\big[y^C_{im} \mid X_i\big] \,.$$

This restriction states that, conditional on municipal characteristics, House electoral performance should contain no additional systematic information about CITREP outcomes. Systematic departures from this condition indicate a breach of the insulation claim embedded in the institutional design.

To operationalize this restriction, we estimate a benchmark linear regression of CITREP vote shares on House vote shares and the vector X_i. The regression is used solely as a computational device to construct residuals and not as an inferential model. Let $\widehat{\mathbb{E}}\big[y^C_{im} \mid y^H_{ij}, X_i\big]$ denote the estimated conditional expectation from this benchmark. The corresponding residuals are given by:

$$\hat{\varepsilon}_{im} = y^C_{im} - \widehat{\mathbb{E}}\big[y^C_{im} \mid y^H_{ij}, X_i\big] \,.$$

These residuals capture deviations from the level of insulation implied by the design. Under correct specification of X_i, sampling variation and measurement error generate deviations that are mean-zero and symmetrically distributed. By contrast, systematic spillover from concurrent House elections induces directional deviations in which CITREP outcomes move closer to House electoral performance than municipal fundamentals alone would predict. The magnitude of $\hat{\varepsilon}_{im}$ therefore serves as a diagnostic statistic for violations of the insulation restriction.

We translate this diagnostic statistic into a binary indicator of influence. Municipality–pair observations in which the absolute value of the residual exceeds a conservative threshold are classified as exhibiting influence, while those below the threshold are classified as insulated. The threshold is chosen to prioritize specificity over sensitivity, ensuring that only substantively meaningful deviations from the insulation condition are labeled. This procedure yields an initial set of municipalities flagged as potentially influenced. Pair-level influence flags are aggregated to the municipality level such that a municipality is classified as influenced if at least one CITREP–House pairing exhibits a substantively meaningful breach of insulation.

To assess the plausibility of these statistical labels, we conduct a validation exercise grounded in observable patterns of electoral mobilization. For each CITREP candidate, we compute two concentration-based metrics: (i) the percentage of the candidate's total votes concentrated in a single municipality, and (ii) the candidate's vote margin within that municipality. Using an interquartile range criterion, we identify candidates exhibiting unusually high levels of territorial concentration along either dimension. These concentration flags are then cross-referenced with the influence labels derived from the residual-based diagnostic.

This validation exercise shows that 69.33% of municipalities initially classified as influenced also exhibit strong territorial vote concentration, consistent with localized patterns of electoral mobilization. This correspondence indicates that the anomalous comovement captured by the diagnostic baseline is not merely a numerical artifact, but aligns with geographically plausible patterns of support. For the remaining cases, where statistical deviations lack clear territorial coherence, we conduct a targeted review and reclassify them conservatively. The purpose of this step is not to refine predictive performance, but to reduce false positives and ensure that influence labels reflect both statistical deviation and contextual feasibility.

The result of this procedure is a municipality-level binary indicator identifying where the insulation condition embedded in the CITREP design plausibly fails. This indicator serves as the outcome for subsequent stages of analysis, which focus on characterizing the territorial and organizational environments in which such deviations from insulation are more likely to arise.

To avoid inviting method-centric detours at this stage, the full specification, data-structure details (including the municipality–combination unit), and diagnostics can be found in Appendix A, along with coefficient tables. Formal definitions and construction of covariates (e.g., municipal performance and intergovernmental transfers) appear in the Data section. Because repeating municipalities across many CITREP–House

pairings could inflate precision, Appendix A reports the baseline estimates underlying this diagnostic test; all results are consistent with a systematic though modest comovement between House and CITREP tallies.

Conceptually, the residual-based labeling procedure described above can be understood as a form of anomaly detection. The insulation restriction defines a normative baseline of electoral independence; systematic deviations from this baseline constitute anomalous comovement between CITREP and House outcomes. In settings where the event of interest is rare and embedded in complex, high-dimensional environments, the anomaly-detection literature emphasizes the importance of flexible, nonparametric approaches capable of identifying structurally meaningful deviations without imposing restrictive functional forms (Czerny, Leitner, et al. 2020; Wen et al. 2019; Zhao, Hryniewicki, Li, et al. 2018). These approaches are particularly relevant when labels reflect departures from an expected baseline rather than direct observation of underlying mechanisms. Consistent with this perspective, we treat residual-based influence flags as detected anomalies relative to the insulation benchmark. The subsequent Random Forest model does not generate these labels; instead, it approximates the latent function mapping municipal characteristics to anomalous insulation failures, allowing us to recover the environmental regularities associated with already identified deviations.

5.3 Characterizing Influence: Random Forest Classification

Having identified municipality–pair observations in which the insulation condition plausibly fails, we next characterize the municipal features that systematically distinguish influenced from insulated cases. The purpose of this step is not prediction for its own sake, but to identify patterns of interaction and nonlinearity that are difficult to specify ex ante and that are central to the notion of influence ecologies developed in the conceptual framework.

We employ a Random Forest classifier trained on the binary influence labels constructed in the diagnostic baseline. Random Forests are well suited to this task for three reasons. First, influence is likely to arise from combinations of municipal characteristics rather than from single factors operating additively. Second, the relationships between these characteristics and influence may be nonlinear and involve threshold effects. Third, the method is robust to multicollinearity and does not require strong parametric assumptions about functional form. These properties make Random Forests appropriate for exploratory characterization in settings where theory motivates configurations rather than isolated covariates.

The classifier is trained using municipal-level covariates measured prior to the 2022 elections. These covariates capture baseline structural conditions common to both electoral contests and reflect dimensions emphasized in the conceptual framework, including state capacity, territorial exposure to armed conflict, and the organizational environment in which CITREP candidates operate. The outcome variable is the binary indicator of influence derived in Section 5.2. Model tuning and validation follow standard practices and are reported in the Appendix.

The Random Forest analysis thus serves a strictly intermediate role. It identifies which municipal features, alone and in combination, are most informative for distinguishing influenced from insulated cases. It does not yield a typology, nor does it establish causal effects. Instead, it provides the empirical foundation for the subsequent latent class analysis, which translates these high-dimensional patterns into a small number of interpretable influence ecologies.

Formally, let $I_i \in \{0, 1\}$ denote the residual-based indicator of partisan influence for municipality i, and let Z_i denote the vector of municipal attributes capturing state capacity, political competition, socioeconomic conditions, and conflict legacies. We conceptualize influence as arising from an unknown, potentially nonlinear mapping

$$\mathbb{P}(I_i = 1 \mid Z_i) = f(Z_i),$$

where $f(\cdot)$ summarizes the structural conditions under which electoral insulation is likely to fail. The objective of the classification model is not to recover causal effects of individual covariates, but to approximate this latent influence function and identify the configurations of municipal characteristics associated with higher predicted probabilities of influence.

Random Forests provide a flexible, nonparametric approximation to $f(\cdot)$ by aggregating predictions over an ensemble of decorrelated decision trees, each of which partitions the covariate space into locally homogeneous regions. This architecture allows the model to capture higher-order interactions and threshold effects among municipal characteristics that are difficult to specify in parametric form.

To interpret the fitted classifier, we rely on Shapley Additive Explanations (SHAP), which decompose each municipality-specific prediction into additive feature contributions. For a given observation i, the predicted log-odds of influence can be written as

$$\hat{f}(Z_i) = \phi_0 + \sum_{k=1}^{K} \phi_k(Z_{ik}),$$

where ϕ_0 is the average prediction over the sample and $\phi_k(Z_{ik})$ denotes the Shapley value associated with covariate k. Each Shapley value represents the average marginal contribution of that feature to the prediction, computed over all possible orderings of covariates. Importantly, SHAP values are not causal effects. They provide a locally additive representation of a global prediction function, allowing us to recover interpretable profiles of municipal environments associated with higher or lower predicted influence.

Model performance is assessed through cross-validation, focusing on precision, recall, and overall accuracy. Beyond predictive accuracy, we employ SHAP values to interpret the contribution of individual variables to classification outcomes. This provides municipality-specific explanations, allowing us to identify not only which features are most predictive on average, but also how they shape predictions in particular local contexts. The result is an interpretable machine-learning framework that links structural conditions to the persistence of partisan influence in CITREP elections.

5.4 From Predictors to Ecologies: Latent Class Analysis

The Random Forest classifier identifies which municipal features distinguish influenced from insulated cases, but it does not reveal *how* those features combine into coherent territorial regimes. To recover interpretable configurations, we employ Latent Class Analysis (LCA) to cluster municipalities into a small number of discrete influence ecologies. This step completes the diagnostic pipeline by translating the high-dimensional patterns recovered by supervised learning into a typology grounded in the configurational logic developed in Section 4.

The LCA is not a second classifier and does not seek to outperform the RF in predictive accuracy. Instead, it summarizes the joint distribution of key municipal features into discrete, interpretable ecologies that illuminate how distinct local environments condition the relationship between traditional politicians and victim-backed CITREP candidates. Where the RF provides a ranked map of variable importance, the LCA reveals the structural alignments through which those variables combine to produce influence. The two approaches are therefore complementary: the RF identifies what matters most on average; the LCA clarifies the configurations in which those factors co-occur.

We implement the LCA using a set of municipal covariates identified as relevant by both our theoretical framework and the RF model. These include: (i) electoral co-movement between House and CITREP tallies (captured by residual-based measures from the diagnostic baseline), (ii) state capacity and administrative resources (e.g., municipal performance index, fiscal balance, central government transfers), (iii) socioeconomic vulnerability and scale (e.g., rurality, poverty indices, population size), (iv) conflict and illicit-economy exposure (e.g., forced displacement, terrorist acts, coca cultivation), and (v) organizational and identity structure (e.g., representation by community councils, ethnic organizations, and social movements). This mapping ensures that the LCA operationalizes the same dimensions emphasized in the conceptual framework, allowing the empirical typology to speak directly to the theoretical expectations articulated in Section 4.

The LCA models observed municipal characteristics as arising from a finite mixture of latent classes, each representing a distinct influence ecology. The model assumes that, conditional on class membership, observed indicators are locally independent. Municipalities are assigned to classes based on maximum posterior probability. Appendix B provides details about the LCA analysis.

We estimate the LCA using maximum likelihood via the Expectation-Maximization (EM) algorithm, implemented in the `poLCA` package for R (Linzer and Lewis 2011). To guard against convergence to local maxima, we run the model with 500 random starting values and retain the solution with the highest log-likelihood. Class enumeration—the choice of K—balances statistical fit with substantive interpretability. We compare models with two through eight classes using the Bayesian Information Criterion (BIC), the sample-size adjusted BIC (aBIC), and the Bootstrap Likelihood Ratio Test (BLRT), alongside entropy as a measure of classification certainty (Nylund-Gibson and Choi

2018). Lower BIC and aBIC values indicate better fit, penalized for model complexity; significant BLRT tests favor the more complex model; and entropy values closer to 1.0 indicate clearer class separation.

We report the six-class solution ($K = 6$) because it provides a distinctive and policy-relevant typology that aligns closely with the municipal profiles identified by the Random Forest while maintaining strong classification quality. A more parsimonious four-class specification was rejected on both empirical and substantive grounds: it produced higher BIC and aBIC values and collapsed analytically meaningful variation across intermediate cases, obscuring distinctions between, for instance, institutionalized enclaves and mobilized frontiers. Conversely, solutions with seven or eight classes introduced overfitting and generated classes with unstable membership and ambiguous substantive interpretation. The six-class model achieves an entropy of 0.89, indicating clear class separation, and average posterior probabilities across classes range from 0.82 to 0.94, confirming that municipalities are confidently assigned. Full model fit statistics, including BIC comparisons across class specifications, are reported in Appendix B.4.

The resulting typology identifies six recurring municipal environments—*influence ecologies*—that structure the conditions under which CITREP insulation succeeds, partially holds, or fails. These ecologies are not deterministic categories but probabilistic configurations, reflecting the most likely alignment of state capacity, organizational density, conflict legacies, and political competition observed in each cluster. Class labels are descriptive shorthand supported by consistent patterns in standardized loadings and convergence with the RF's SHAP-based explanations. We use them as interpretive devices to structure the substantive discussion in Section 7 rather than as fixed categories. This framing preserves the analytical nuance of the typology and aligns with the broader goal of diagnosing structural variation in CITREP outcomes without treating territorial regimes as fixed or natural categories.

Table 1 presents the core latent class profiles underlying the six-class solution. To preserve interpretability and maintain coherence with the preceding Random Forest analysis, the table reports a restricted set of *core predictors*—those identified as most influential in the SHAP-based importance rankings. All entries correspond to standardized mean values within each latent class, while the final column reports the relative size (*share*) of each class.

The profiles reveal clear and substantively meaningful contrasts across influence ecologies. Variation in coalition size, organizational representation within CITREP lists, rural population share, and educational scale jointly structures the municipal environments in which insulation from traditional political influence succeeds, partially erodes, or fails. Importantly, no single variable defines any class in isolation. Instead, each ecology is characterized by a distinctive *configuration* of political organization, territorial context, and demographic scale, consistent with the configurational logic developed in Section 4.

While Table 1 highlights the most theoretically and empirically diagnostic dimensions of the latent typology, the full set of standardized class profiles—including all municipal covariates used in the estimation—is reported in Appendix D.

Table 1: Latent class profiles for core predictors

Latent class	House coalition size	CITREP Social movements	CITREP Displaced	CITREP Indigenous	CITREP Productive sector	Rural population share	CITREP Women	Population per school	share
0	-0.002	-0.011	0.002	0.008	0.003	0.013	0.000	0.003	0.287
1	0.009	-0.007	-0.006	0.004	0.002	0.009	0.000	0.004	0.132
2	-0.003	-0.012	-0.002	0.000	0.004	-0.013	0.004	0.003	0.246
3	0.001	-0.008	0.016	0.016	0.010	0.023	-0.002	0.005	0.042
4	0.005	-0.008	-0.005	0.001	-0.001	0.016	0.013	0.008	0.096
5	0.005	-0.007	-0.003	0.001	0.003	-0.018	0.002	-0.004	0.198

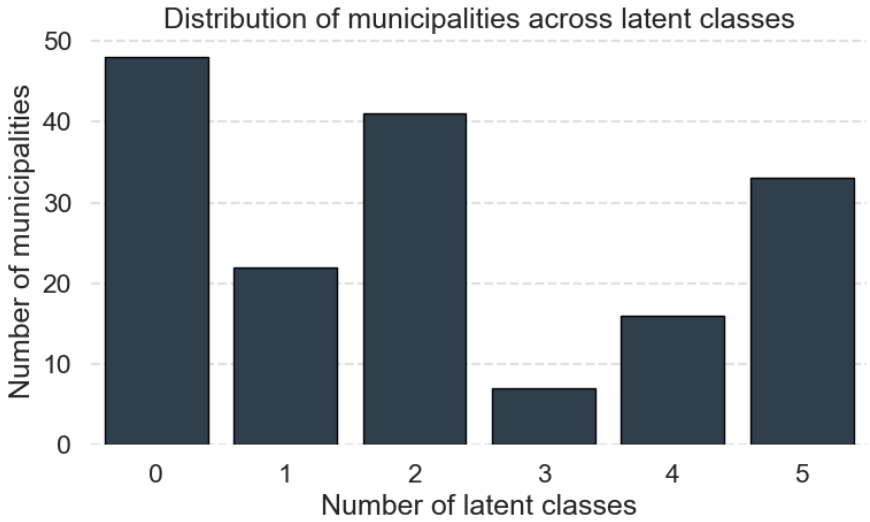

Figure 2: Distribution of municipalities across latent classes

Figure 2 reports the distribution of municipalities across the six latent classes identified by the LCA. Class sizes are uneven, reflecting substantive heterogeneity rather than model artifacts. Two classes (Classes 0 and 2) account for a large share of municipalities, capturing the most common institutional and political environments. At the other extreme, Class 3 is relatively small, indicating a more specific and less prevalent configuration of municipal characteristics. This dispersion underscores that the latent typology does not merely partition municipalities mechanically, but instead recovers distinct influence ecologies that vary meaningfully in prevalence and territorial reach.

The LCA thus completes the empirical strategy by providing a disciplined bridge from the RF's high-dimensional feature space to a small set of substantively meaningful environments. Together, these stages—diagnostic baseline, Random Forest classification, and latent class analysis—form a coherent pipeline for detecting and characterizing traditional political influence in Colombia's peace constituencies. Section 6

presents the results of this analysis, beginning with the diagnostic patterns of influence and culminating in the six-class typology and its spatial distribution across CITREP municipalities.

6 Results

This section presents the empirical findings from the three-stage diagnostic pipeline. We proceed sequentially, moving from the baseline patterns of anomalous comovement to the structural predictors identified by the Random Forest classifier, and finally to the six influence ecologies recovered through Latent Class Analysis. Each subsection is descriptive rather than inferential. The goal is not to test causal hypotheses but to characterize the territorial and organizational environments in which the insulation condition embedded in the CITREP design plausibly fails.

6.1 Diagnostic Patterns of Influence

The residual-based labeling procedure described in Section 5.1 identifies systematic departures from the insulation restriction across CITREP municipalities. The diagnostic baseline regression—estimated at the municipality-pair level and aggregated to the municipal level—reveals that anomalous comovement between CITREP and concurrent House of Representatives vote shares is neither rare nor uniformly distributed. Of the 167 municipalities in which CITREP elections were held, a substantial proportion exhibit residual deviations that exceed the conservative threshold set to prioritize specificity over sensitivity. These municipalities are classified as potentially influenced, reflecting cases in which CITREP electoral outcomes moved closer to House performance than municipal fundamentals alone would predict.

The residual-based labeling procedure identifies systematic departures from the insulation restriction across CITREP municipalities. Of the 167 municipalities in which CITREP elections were held, a substantial proportion exhibit residual deviations that exceed the conservative threshold set to prioritize specificity over sensitivity.

The spatial distribution of influence reveals marked territorial heterogeneity. Municipalities flagged as influenced are not randomly scattered but cluster in recognizable corridors. Institutionalized High-Conflict Enclaves appear across fifteen departments, including portions of the Caribbean interior (Bolívar, Cesar, Magdalena), though this region exhibits substantial internal variation mixing high, moderate, and low-influence environments. Putumayo displays comparatively lower prevalence of influence (87.5% Expanding Community Councils), reflecting environments where organizational density and community-led intermediation provide greater insulation. Cauca and Nariño exhibit pronounced heterogeneity, hosting four distinct ecology types each. These patterns are not reducible to departmental boundaries; influence ecologies cut across administrative divisions, suggesting that mechanisms sustaining or resisting partisan penetration operate at the municipal rather than regional scale.

Importantly, the prevalence of influence varies not only across space but also across the sixteen CITREP electoral districts themselves. Because district boundaries were designed to reflect conflict geographies rather than administrative coherence, many districts combine municipalities with divergent structural characteristics and organizational capacities. Some districts aggregate municipalities where influence is systematically detected alongside others where insulation appears to hold, producing compositional heterogeneity that complicates any district-level assessment of CITREP performance. This internal diversity underscores a central design challenge: the territorial logic of the peace constituencies assumed conflict-affected municipalities would share similar representational needs, but the empirical record reveals profound variation in the feasibility of autonomous political mobilization even within narrowly defined conflict zones.

The diagnostic baseline thus establishes two core empirical regularities. First, the insulation condition embedded in the CITREP design does not hold uniformly; systematic breaches of electoral independence are detectable and territorially patterned. Second, these patterns align with observable features of electoral mobilization—candidate vote concentration, spatial clustering, and district composition—rather than statistical noise. Together, these findings motivate the subsequent stages of analysis, which seek to characterize the municipal environments in which anomalous comovement is most likely to arise.

6.2 Predictors of Influence: Random Forest Analysis

The Random Forest classifier trained on the binary influence labels identifies a sharply hierarchical structure of predictors. The ranked map of variable importance, summarized through mean absolute SHAP values, reveals which municipal characteristics most consistently shape the probability that a municipality exhibits anomalous comovement between CITREP and House electoral outcomes. SHAP values quantify the average contribution of each feature to the model's predictions across all municipalities, providing a comparative measure of explanatory importance without imposing causal interpretation. Higher SHAP magnitudes indicate that a variable consistently shifts predicted probabilities, while lower values exert only marginal effects.

Table 2 presents the top ten predictors ranked by mean absolute SHAP value. Coalition size—the breadth and cohesion of traditional party alliances in concurrent House elections—emerges as the single most influential feature, with a SHAP value of 0.085.[1] This finding underscores a structural continuity between Colombia's formal party system and its supposedly insulated peace constituencies: where professional political networks fielded broad coalitions and maintained embedded territorial machinery, they possessed the mobilizational capacity to project influence into adjacent CITREP districts without overtly violating exclusion rules. The dominance of coalition size confirms that traditional political influence operates primarily through organiza-

[1] The complete ranking of all 46 predictors is visualized in Appendix **??**.

tional leverage rather than through economic coercion or administrative absence.

Table 2: Top Ten Predictors of Influence by Mean Absolute SHAP Value

Rank	Variable	Mean SHAP
1	House coalition size	0.0854
2	CITREP - Social movements	0.0752
3	CITREP - Victims' organizations	0.0659
4	CITREP - Displaced populations	0.0573
5	CITREP - Indigenous communities	0.0540
6	CITREP - Productive sector	0.0475
7	Rural population share	0.0408
8	CITREP - Women's organizations	0.0294
9	Population per school	0.0270
10	CITREP - Afro-descendant councils	0.0242

Immediately below coalition size sits a dense cluster of variables corresponding to different types of victim movements. These organizational categories—social movements, victims' organizations, displaced populations, indigenous communities, productive sector associations, women's organizations, and Afro-descendant community councils—collectively account for the majority of the model's core predictive structure. Their combined SHAP contribution, alongside coalition size, accounts for more than half of the model's core predictive structure. This organizational variation reveals that the permeability of institutional firewalls is determined less by the formal exclusion of parties than by the uneven organizational capacity of victim movements themselves.

These organizational categories vary considerably. At one end are broad, socially embedded associations such as social movements and federated peasant organizations, which possess horizontal networks, cross-municipal coordination, and prior engagement with state and NGO actors. These organizations function as partial substitutes for state capacity, providing autonomous mobilization channels that shield CITREP candidates from external co-optation. Their presence in the predictive hierarchy with substantial SHAP values indicates that where such movements are active, predicted influence declines. At the other end are narrow, identity-specific or economically dependent organizations—productive sector associations embedded in patronage-rich commodity economies, or territorially bounded indigenous movements reliant on external alliances for electoral resources. Where these organizational forms dominate, predicted influence rises, reflecting structural vulnerabilities that traditional brokers can exploit through alliances of convenience.

Between these poles lie intermediate organizational types. Victims' organizations and displaced populations' movements hold moral legitimacy but often lack stable funding and technical capacity, making them dependent on local sponsorship. Indigenous movements are territorially anchored within cabildos that provide community authority but face resource constraints when contesting elections beyond their jurisdictions.

Women's organizations and Afro-descendant community councils occupy mixed positions: where federated and linked across municipalities, they preserve CITREP independence; where fragmented or resource-poor, they provide institutional shells through which traditional actors can reenter. This organizational heterogeneity, visible in the distribution of SHAP values across victim movement categories, underscores that "victim representation" is not a monolithic category but a spectrum of organizational forms with vastly different capacities for autonomous mobilization.

Two municipal-structural variables complete the top ten and anchor the environmental context in which these political dynamics unfold. Rural population share (SHAP 0.041) captures the geography of state reach and monitoring capacity. Higher rurality increases predicted influence, reflecting the logistical and informational asymmetries characteristic of dispersed populations where oversight is limited and personalistic brokerage substitutes for programmatic competition. Population per school (SHAP 0.027) measures the density of public education infrastructure, serving as a proxy for everyday state presence and civic socialization. High ratios—few schools per inhabitant—signal institutional thinness, and the positive SHAP contribution indicates that where educational infrastructure is scarce, the state's routine visibility declines, leaving representational voids that brokers fill.

Beyond the top ten, a second tier of predictors refines the model's geographic and socioeconomic resolution. Variables measuring service provision—subsidized health enrollment (SHAP 0.024), garbage collection coverage (SHAP 0.018), access to clean water (SHAP 0.018)—differentiate municipalities with universal, impersonal service delivery from those where access depends on administrative mediation. Measures of poverty, inequality, and forced displacement delineate environments of vulnerability in which traditional brokers retain leverage. Spatial variables such as distance to urban centers (SHAP 0.018) and distance to markets (SHAP 0.018) mark the informational and monitoring costs that modulate how political networks operate. Collectively, these Tier 2 variables account for an additional 21% of total model importance, providing the socioeconomic and geographic scaffolding for the organizational patterns identified in Tier 1.

Conflict-related variables—coca cultivation (SHAP 0.019), forced displacement (SHAP 0.019), homicides (SHAP 0.018), landmine victims (SHAP 0.015), child recruitment (SHAP 0.015), terrorism incidents (SHAP 0.009)—appear throughout the hierarchy but do not dominate predictive importance. Their modest SHAP contributions indicate that violence and illicit economies condition the environments in which influence operates but do not drive influence directly. Instead, they amplify the mechanisms identified above: in municipalities with high displacement or coca cultivation, the logistical and economic barriers to autonomous organization rise, heightening the advantages of traditional brokers who possess established networks. In this sense, conflict exposure serves as a structural vulnerability that magnifies organizational asymmetries rather than an independent causal mechanism.

Administrative capacity variables—municipal performance index (SHAP 0.014), fiscal performance index (SHAP 0.011), administrative management index (SHAP

0.015)—also rank in the middle to lower tiers, confirming that state capacity alone does not guarantee insulation. Where administrative efficacy coexists with hybrid governance orders, capable local states can be instrumentalized by political machines, transforming efficient service delivery into an intrument of clientelistic control. This pattern, fully visible only when the Random Forest results are interpreted alongside the Latent Class Analysis, resolves the apparent paradox of high predicted influence in administratively capable municipalities.

The cumulative distribution of SHAP importances reveals a strong concentration: the top eighteen variables explain two-thirds of total model variation, while features below SHAP 0.017 form a long tail of minor predictors. This hierarchical structure confirms that influence in post-conflict electoral institutions arises from a relatively compact set of organizational and territorial conditions rather than diffuse multicausality. The Random Forest analysis thus provides a ranked map of predictive importance that foregrounds organizational variables—coalition size and victim movement types—as the primary drivers of influence, with municipal structure, service provision, and conflict exposure serving as contextual modifiers that refine classifications in specific local environments.

Importantly, these patterns emerge from a nonparametric learning algorithm designed to capture interactions and nonlinearities that escape conventional regression models. The SHAP framework allows us to decompose predictions into additive feature contributions at the municipality level, providing interpretable profiles of the structural conditions associated with higher or lower predicted influence. The result is an empirically grounded characterization of how traditional political networks penetrate spaces meant for autonomous victim representation, revealing systematic asymmetries in organizational capacity that the formal design of the CITREP failed to anticipate or address.

6.3 Six Influence Ecologies

Configurational logic anticipates that municipalities do not array along a single insulation-capture continuum, but cluster into discrete ecology types. Where state capacity is high and organizational density concentrated, community institutions should exercise gatekeeping functions, insulating CITREP from traditional influence. Where state capacity is high but victim organizations fragmented, administrative resources provide pathways for co-optation by capable machines. Where both state capacity and organizational density are low, neither autonomous movements nor traditional parties possess sufficient infrastructure to compete effectively, leaving seats vulnerable to opportunistic capture or non-participation. The question is whether theoretically implied configurations emerge empirically, or whether municipalities exhibit continuous variation that resists typological classification.

The six-class LCA solution reported in Section **??** confirms that municipalities cluster into discrete configurations rather than distributing continuously across predictor space. With entropy of 0.89 and average posterior probabilities exceeding 0.82 across

all classes, the typology recovers sharp boundaries between ecological types. Importantly, the solution is not data-mined: the six-class specification outperforms more parsimonious alternatives on both statistical criteria (lower BIC/aBIC) and substantive interpretability (preserves theoretically meaningful distinctions such as Quiet vs. Deprived Peripheries, Depoliticized Semi-Centers vs. Mobilized Frontiers), while more complex seven- or eight-class solutions produce unstable membership and ambiguous interpretation. The recovered ecologies align with theoretical patterns while revealing finer-grained distinctions—such as the division between mobilized and depoliticized intermediate zones—that could not be deduced from theory alone. Full class profiles across all municipal covariates—organized by health/education, public services, organizational representation, political organization, and conflict/territory dimensions—are reported in Appendix D. What follows describes each ecology and its relationship to CITREP insulation.

The Latent Class Analysis translates the high-dimensional predictive patterns recovered by the Random Forest into a small number of interpretable influence ecologies. These ecologies represent recurring configurations of municipal characteristics that jointly condition the feasibility of electoral insulation. The LCA does not seek to outperform the Random Forest in predictive accuracy; instead, it summarizes the joint distribution of key municipal features into discrete, substantively meaningful environments that clarify how distinct local contexts shape the relationship between traditional politicians and victim-backed CITREP candidates.

The Latent Class Analysis identifies six distinct influence ecologies based on the joint distribution of municipal characteristics (model selection criteria reported in Appendix B). Table 3 summarizes the six influence ecologies, reporting class prevalence and defining characteristics across key dimensions of state capacity, organizational density, conflict exposure, and predicted influence. Class labels are descriptive shorthand supported by consistent patterns in standardized loadings and by convergence with the Random Forest's SHAP-based explanations. We use them as interpretive devices to structure the substantive discussion rather than as fixed natural categories.

Table 3: Six Influence Ecologies: Class Characteristics

Ecology	**N**	**%**	**State Capacity**	**Org. Density**	**Conflict Exposure**	**Pred. Influence**
Quiet Peripheries	28	16.8	Low	Low	Low	Low
Expanding Councils	23	13.8	Moderate	High	Low	Very Low
Depoliticized Semi-Centers	31	18.6	Moderate	Low	Low	Moderate
Mobilized Frontiers	29	17.4	Moderate	Moderate	High	High
Institutionalized Enclaves	34	20.4	High	Low	High	Very High
Deprived Peripheries	22	13.2	Very Low	Low	Low	Low
Total	**167**	**100.0**				

The spatial distribution of these six ecologies, shown in Figure 3, reveals a politically fragmented landscape where influence patterns cut across departmental bound-

aries. Some classes concentrate in recognizable geographic corridors—Mobilized Frontiers in Norte de Santander, Expanding Community Councils in the Pacific-Amazonian arc—while others, particularly Institutionalized High-Conflict Enclaves and Quiet Peripheries, appear dispersed across multiple regions. This municipal-level heterogeneity confirms that the mechanisms sustaining or resisting partisan penetration operate at the local rather than regional scale. The following paragraphs characterize each ecology in turn, highlighting defining features and geographic distribution.

Quiet Peripheries sit at the low end of administrative and fiscal resources and exhibit low levels of violence relative to the rest of the sample. Education attainment, central government transfers, revenues, and administrative staffing fall below the cross-municipal average, as do indicators of armed conflict such as kidnappings, forced disappearances, and landmine victimization. These are rural peripheries where the conflict either waned or never reached high intensity, but where the state's managerial footprint remains thin. The pattern is not one of institutional strength pacifying the territory but quiet born of disconnection. In this ecology, predicted influence is low, but this does not signal CITREP autonomy guaranteed by robust local counterweights. Rather, it reflects that disconnection from both conflict and mobilization renders these municipalities unattractive targets for partisan penetration. The absence of influence signals not autonomous representation but a representational vacuum where neither victim organizations nor traditional machines possess the resources to compete effectively. Quiet Peripheries are dispersed across ten departments, including the Andean southwest, the Amazon frontier, and portions of the Bajo Cauca, reflecting a structural condition of institutional thinness that manifests independently of regional identity.

Expanding Community Councils are characterized by territorially anchored community organization, most notably Afro-descendant community councils, paired with relatively large candidate coalitions and a muted imprint of traditional ideological camps. Displacement and forced disappearance rates are comparatively low, and distance to markets is not especially high. This ecology is consistent with bottom-up modes of representation where organizational density and social authority are supplied not by partisan brokers but by community institutions with historical roots and internal accountability mechanisms. In such settings, the CITREP design operates closer to its normative intent: it channels victim representation through non-partisan networks that possess the capacity to resist external co-optation. Predicted influence in this ecology is very low, reflecting that where communal structures are present and municipal capacity is at least moderate, insulation proves viable. Geographically, Expanding Community Councils are concentrated in Putumayo (87.5% of its CITREP municipalities) and scattered across the Pacific-Amazonian corridor, including portions of Chocó, Cauca, and Nariño, where ethnic federalism and territorial autonomy provide pre-existing social infrastructures that reinforce institutional firewalls.

Depoliticized Semi-Centers gather small to mid-sized semi-urban localities that are somewhat better off in education, revenues, and transfers and are less rural than the modal CITREP municipality. They display weak ideological alignment and thin organizational layering across the social categories observed in the data. Service coverage

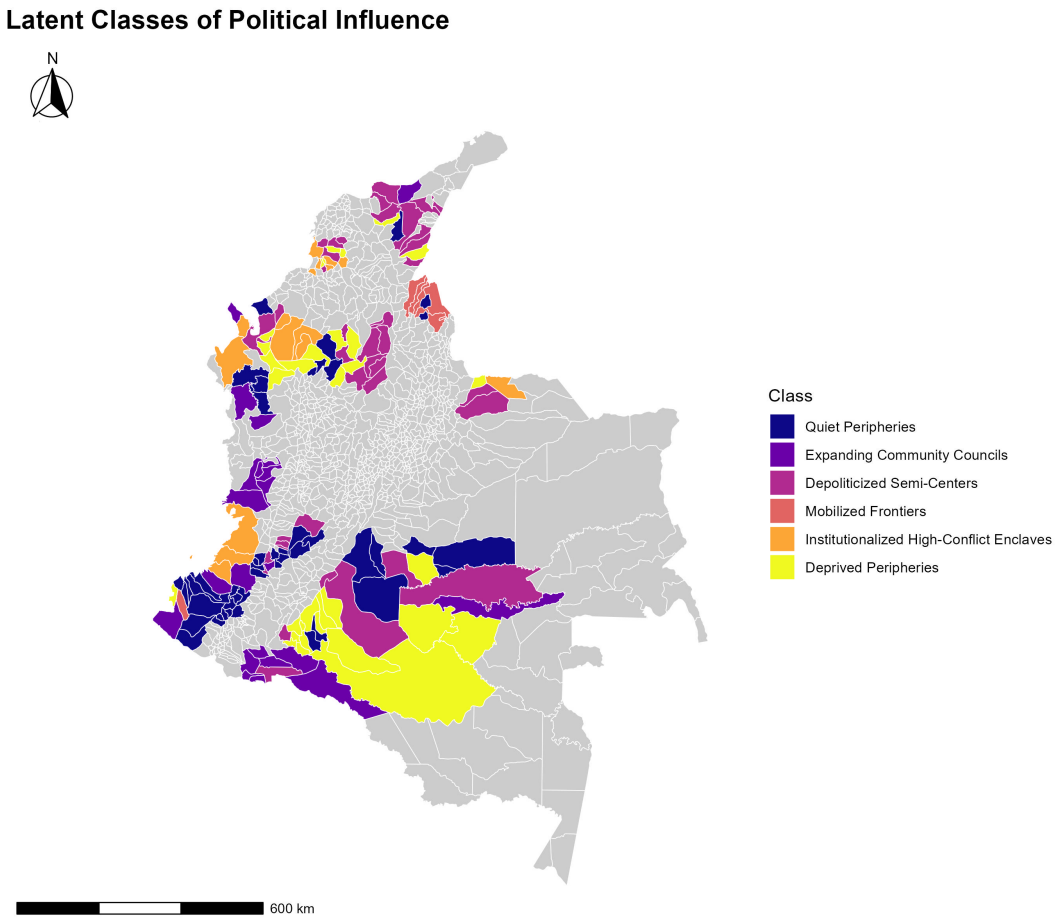

Figure 3: *Spatial distribution of latent classes of political influence across CITREP municipalities. The map shows that influence ecologies are not confined to departmental borders: some classes cluster in specific regions—such as the Pacific–Amazonian corridor or the Caribbean interior—while others are dispersed nationwide. This spatial heterogeneity underscores that patterns of traditional political influence vary primarily at the municipal, rather than departmental, level.*

is not uniformly strong, but basic administrative routines are in place. Conflict exposure measures are comparatively modest. This ecology points to coordination through technocratic channels rather than through parties or movements. Predicted influence is moderate, lower than in the conflictual classes but higher than in the communal class, precisely because there are fewer endogenous social counterweights to buffer against external mobilizers. In these environments, low politicization is not the same as autonomy; it can also signal an open lane for professionalized political networks to fill the representational void. Depoliticized Semi-Centers are present in Antioquia and scattered across the Caribbean interior (Bolívar, Cesar, Magdalena), reflecting semi-urban nodes that bridge Caribbean strongholds, Andean corridors, and peripheral mining zones.

Mobilized Frontiers are high-mobility, economically mixed territories—frontiers where licit and illicit economies intersect. They stand out for above-average displacement and coca cultivation, longer distances to markets, higher poverty, and the emergence of new institutional structures and service coverage reflecting recent state-building efforts under the PDET framework. These are not voids; they are sites of active reconfiguration where victim organizations, armed governance legacies, and traditional brokers compete for influence. Predicted influence is high, reflecting that the same features powering partisan penetration—mobile populations, resource flows, limited oversight capacity—also sustain collective mobilization by social movements and community actors. This ecology captures the intuitive but hard-to-measure possibility that CITREP becomes an arena of contestation where the outcome remains endpoint-sensitive: small shifts in oversight, organizational capacity, or resource flows can tip the equilibrium toward autonomy or capture. Mobilized Frontiers are sharply localized in Norte de Santander, where they account for 75% of CITREP municipalities, reflecting a frontier-specific ecology shaped by cross-border coca economies and the recent arrival of post-accord infrastructure.

Institutionalized High-Conflict Enclaves are the largest and most consequential class, representing one-fifth of all CITREP municipalities. Here, indicators of coercion—kidnappings, terrorist acts, enforced disappearances—and forced displacement are high, but so are fiscal performance, administrative staffing, and coverage of core services such as water access and subsidized health enrollment. Municipal revenues and expenditures are stronger than in the average CITREP municipality. The combination suggests not state absence but coexistence: the state is present and operational, yet governance is negotiated in ways that allow violent entrepreneurship and political clientelism to endure. Predicted influence in this ecology is very high, reflecting that the problem is not that CITREP lacks an institutional partner on the ground but that the partner is often embedded in hybrid orders where efficient administration and machine politics function as complements rather than substitutes. This class makes visible a sophisticated form of co-optation—one that thrives in "working" local states capable of delivering services while simultaneously channeling political mobilization through clientelistic networks. Institutionalized High-Conflict Enclaves are transversal across fifteen of nineteen departments, stretching from the Caribbean interior (Bolívar,

Cesar, Magdalena) through Antioquia's Bajo Cauca and Urabá into isolated pockets in Meta and Cauca. Their geographic dispersal indicates that the conditions sustaining sophisticated co-optation recur wherever state capacity coexists with entrenched clientelism and residual armed governance.

Deprived Peripheries group municipalities with consistently low coverage of basic services—sanitation, sewerage, waste collection—low central government transfers, weak fiscal and administrative baselines, and, strikingly, low incidence of open conflict. They are the late developers of the CITREP universe: poor, rural, and thinly governed, but not necessarily violent. Predicted influence is low, not because insulation succeeds through robust counterweights but because without resources and organizational density, traditional machines have fewer short-term levers to pull. The absence of influence does not imply a self-protecting equilibrium; it indicates that autonomous mobilization is more than a formal possibility requiring substantial reduction in the fixed costs of participation. Deprived Peripheries are dispersed across ten departments, including portions of Caquetá (75% of its CITREP municipalities), the Amazon frontier, and the northern Antioquia, reflecting environments where representational promise remains structurally unfulfilled.

The spatial distribution of these six ecologies reveals three critical patterns. First, influence ecologies cut across departmental boundaries. Institutionalized High-Conflict Enclaves appear in fifteen of nineteen departments; Quiet Peripheries and Deprived Peripheries each span ten departments. This transversal presence confirms that susceptibility to influence cannot be predicted from departmental affiliation; it must be diagnosed at the municipal level. Second, within-department diversity is substantial. Cauca and Nariño host four distinct classes each, exhibiting high entropy indices above 1.0, while Antioquia's twenty-four CITREP municipalities divide nearly evenly among Quiet Peripheries, Depoliticized Semi-Centers, and Deprived Peripheries. By contrast, Putumayo (87.5% Expanding Community Councils), Caquetá (75% Deprived Peripheries), and Norte de Santander (75% Mobilized Frontiers) exhibit low entropy and high concentration, indicating more uniform territorial regimes. Third, many CITREP electoral districts aggregate municipalities from multiple latent classes, creating compositional heterogeneity that complicates district-level assessments. A candidate may secure autonomous support in Expanding Community Councils municipalities through community endorsement while relying on broker-mediated mobilization in Institutionalized High-Conflict Enclaves within the same district.

The six-ecology typology thus provides a rigorous bridge from the Random Forest's high-dimensional feature space to a small set of substantively meaningful environments. Together, the diagnostic baseline, Random Forest classification, and Latent Class Analysis form a coherent pipeline for detecting and characterizing traditional political influence in Colombia's peace constituencies. The typology does not reify territorial regimes as fixed categories but offers an interpretive framework for understanding how state capacity, organizational density, conflict legacies, and political competition combine to make electoral insulation more or less viable across fragmented post-conflict landscapes.

7 Discussion

This paper began from a deceptively simple premise: that Colombia's peace constituencies were designed to insulate victim representation from traditional political influence. The empirical record shows that this insulation was partial, uneven, and territorially contingent. The question is not whether the CITREP succeeded or failed in the aggregate, but *why* identical institutional rules produced radically different outcomes across municipalities. This section interprets those patterns by articulating three core insights: the primacy of the design-territory interaction in conditioning institutional performance, the centrality of configurational rather than additive causation, and the implications of these findings for post-conflict representation in fragmented states.

7.1 The Design-Territory Interaction as Structural Constraint

The six-ecology typology exposes a fundamental design flaw: the CITREP assumed that legal exclusion would be sufficient to protect autonomy, but autonomy is not a rule-based outcome. It is a relational property of local political orders, contingent on the alignment of state visibility, organizational density, and the territorial reach of professional networks. The institutional firewall did not fail uniformly; it failed where territorial conditions rendered it permeable. This is the core meaning of the design-territory interaction: identical rules produce divergent outcomes when inserted into fragmented landscapes of uneven governance. The normative aspiration of political reparation confronted the material reality of subnational heterogeneity, and the latter proved determinative.

7.2 Configurational Logic Over Additive Models

The analytical implication is that influence in post-conflict electoral institutions cannot be modeled additively. It arises from *configurations* of organizational form, state presence, and political competition that vary across municipalities. High state capacity protects insulation when paired with cohesive victim movements but proves insufficient when facing powerful extractive economies and fragmented social organizations. Similarly, organizational fragmentation does not lead to co-optation in municipalities with weak party networks, while even cohesive movements face challenges in environments of extreme administrative scarcity. This configurational logic, anticipated in the conceptual framework and confirmed by the Random Forest's capacity to capture nonlinear interactions, underscores that the ecology of influence is a system property, not a variable effect. The dominance of organizational variables in the predictive hierarchy reveals that influence operates primarily through asymmetries in mobilizational capacity, not through economic coercion or violence.

7.3 Stratified Autonomy in Comparative Perspective

The concept of stratified autonomy connects this study to broader literatures on institutional performance under territorial fragmentation. In federal systems, scholars have long recognized that identical constitutional provisions produce divergent outcomes across subnational units depending on state capacity and party system characteristics (Gibson 2013; Giraudy 2015). In ethnically divided societies, reserved seat systems deliver substantive representation for some groups while enabling co-optation of others, patterned by organizational cohesion and fiscal autonomy (Htun 2004; Krook 2009). In decentralization reforms, identical transfers of authority empower some local governments while creating opportunities for capture in others, contingent on administrative capacity and civil society strength (Falleti 2010; Holland 2017).

Stratified autonomy generalizes across these contexts. It describes the systematic patterning of institutional effectiveness by territorial conditions that formal design cannot erase. The Colombian case demonstrates this logic in a post-conflict setting: CITREP insulation succeeded where community institutions provided autonomous gatekeeping, failed where capable states had been instrumentalized by machines, and proved irrelevant where neither victim organizations nor traditional parties possessed mobilizational capacity. But the mechanism—organizational asymmetries interacting with uneven state presence—recurs wherever inclusionary institutions confront fragmented political landscapes. Understanding stratified autonomy as a general property of institutional performance in heterogeneous settings shifts the design question from "do reserved seats work?" to "where do they work, and what interventions make them work elsewhere?"

7.4 Implications for Post-Conflict Representation

The Colombian case offers sobering lessons for the design of inclusionary institutions in transitional democracies. These lessons travel beyond the specifics of the CITREP to other post-conflict settings deploying reserved seats, special districts, or transitional electoral arrangements in contexts marked by territorial inequality.

7.4.1 Legal Insulation is Necessary but Insufficient

Formal exclusion rules—bans on party endorsements, eligibility restrictions, candidate certification—create the juridical space for autonomous representation, but they do not guarantee it. The permeability of institutional firewalls depends on whether local conditions support independent mobilization, and those conditions vary sharply across subnational units. In contexts marked by uneven state capacity, entrenched clientelism, and legacy networks of armed governance, identical rules will produce divergent outcomes. The normative aspiration of inclusion must therefore be matched by a territorial diagnosis of where autonomy is structurally feasible. Designers cannot assume that institutional architecture alone will protect marginalized groups; they must assess

whether the local ecology sustains the organizational and administrative prerequisites for autonomous politics. This diagnostic imperative applies to other post-conflict reserved seat systems. Regional ethnic autonomy institutions in Mexico protected indigenous communities from narco-governance because they provided pre-existing organizational infrastructure (Ley, Mattiace, and Trejo 2019), while Afro-Colombian reserved seats face similar tensions between group consciousness and institutional performance (Villamizar-Chaparro and Echeverri-Pineda 2025).

7.4.2 Organizational Infrastructure Precedes Representational Autonomy

The single strongest predictor of CITREP insulation was not state capacity, violence, or poverty, but the density and cohesion of victim movements themselves. Where social organizations possessed horizontal coordination, prior advocacy experience, and internal accountability mechanisms, they functioned as independent gatekeepers capable of resisting external co-optation. Where they lacked these resources, traditional brokers filled the representational vacuum. This implies that transitional electoral designs cannot treat victim organizations as pre-constituted political actors with equivalent mobilizational capacity. Building autonomy requires sustained investment in the organizational ecosystems that allow marginalized populations to act collectively—federating local councils, providing technical assistance for campaign finance and legal compliance, securing physical protection for leaders, and lowering the fixed costs of cross-municipal coordination. Without these investments, inclusionary institutions risk producing symbolic inclusion rather than substantive representation.

This organizational prerequisite helps explain divergent outcomes in other post-conflict reserved seat systems. In Mexico, regional ethnic autonomy institutions protected indigenous communities from narco-governance precisely because they provided pre-existing organizational infrastructure independent of criminal and partisan networks (Ley, Mattiace, and Trejo 2019). By contrast, Afghanistan's quota systems for women and minorities struggled because beneficiaries lacked autonomous organizational capacity, rendering them vulnerable to co-optation by warlord networks and traditional elites (Bjelica and Sorush 2018; Felbab-Brown 2021).

7.4.3 State Capacity as Double-Edged Sword

The conventional wisdom holds that stronger local states should protect democratic autonomy by providing public goods and reducing the representational vacuum filled by informal brokers. The evidence partially supports this expectation: in municipalities with moderate administrative capacity and weak political machines, state presence does correlate with lower predicted influence. But the *Institutionalized High-Conflict Enclaves* reveal the limits of this logic. In these municipalities, relatively high fiscal performance, administrative staffing, and service coverage coexist with elevated predicted influence, indicating that capable local states can be instrumentalized by political machines when governance operates through hybrid orders. Efficient service

delivery becomes a complement to clientelism rather than a substitute for it, allowing traditional actors to capture the state's formal administrative efficacy for partisan ends.

The implication is that state-building in post-conflict peripheries must attend not only to administrative capacity but to the *quality* of state-society linkages and the autonomy of local institutions from entrenched networks. This finding resonates with scholarship on co-opted state reconfiguration in Latin America, where criminal organizations and partisan machines exploit administrative infrastructure to consolidate territorial control (Abello-Colak and Guarneros-Meza 2014; Galeano Escucha 2018). It suggests that electoral oversight mechanisms must target not just illegal campaign practices but the deeper structural interfaces between public administration and machine networks—procurement processes, personnel appointments, and the allocation of discretionary resources.

7.4.4 Aggregation Bias and the Territorial Logic of Districts

The spatial distribution of influence ecologies challenges the territorial logic of the CITREP design. The latent class analysis reveals that influence ecologies cut across departmental boundaries and produce profound within-district heterogeneity. Many CITREP electoral districts combine municipalities from multiple latent classes—*Expanding Community Councils* territories where autonomous representation thrives alongside *Institutionalized High-Conflict Enclaves* where machines dominate, or *Mobilized Frontiers* in flux adjacent to *Quiet Peripheries* disconnected from both conflict and mobilization. This compositional complexity means that aggregate CITREP performance metrics inevitably conflate distinct generative processes. A candidate may secure autonomous support in some municipalities through community endorsement while relying on broker-mediated mobilization in others within the same district.

Evaluating the success or failure of peace constituencies at the district level therefore obscures the municipal-level variation that determines whether insulation holds. Future designs should either construct territorially homogeneous districts based on empirical clustering of political-organizational characteristics rather than abstract conflict geographies, or explicitly monitor within-district heterogeneity to prevent the aggregation bias that masks capture. This applies to any multi-municipality electoral district in heterogeneous settings: designers must either accept compositional diversity and implement differentiated oversight, or partition districts to align with actual ecologies rather than administrative convenience.

7.4.5 Differentiated Interventions by Ecology Type

The six influence ecologies identified by the LCA are not merely descriptive categories; they are diagnostic tools that clarify where different interventions are likely to prove effective. A one-size-fits-all approach to electoral oversight ignores the structural diversity of post-conflict environments and risks misallocating scarce enforcement resources.

Instead, interventions should be tailored to the specific mechanisms through which influence operates in each ecology.

In *Expanding Community Councils*, the priority is protecting community-led intermediation. This requires shielding leaders from intimidation, formalizing consultation channels, ensuring that communal endorsements are legally recognized, and preventing external actors from exploiting internal community divisions. Electoral authorities should defer to existing community governance structures rather than imposing external validation criteria that undermine their autonomy. In *Institutionalized High-Conflict Enclaves*, tools must target the interface between public administration and machine networks. Procurement transparency, conflict-of-interest screening for local officials, campaign finance audits focusing on discretionary municipal expenditures, and independent oversight of service delivery contracts address the specific mechanism through which capable states are co-opted. Generic compliance monitoring will fail here because the problem is not rule-breaking but the strategic use of legal administrative authority for partisan ends.

In *Mobilized Frontiers*, oversight must be paired with investments that lower participation costs for victim organizations so they can contest traditional networks on more equal terms. This includes travel stipends for candidates from remote municipalities, security protocols for organizers operating in high-risk environments, and legal support to navigate complex registration and reporting requirements. These are not subsidies but correctives to the structural disadvantages that fragmented movements face when competing against established machines. In *Quiet* and *Deprived Peripheries*, basic state-building—reliable services, modest but predictable transfers, physical infrastructure that reduces isolation—may be a precondition for any representational promise to materialize. Without minimum levels of connectivity and administrative presence, neither autonomous movements nor oversight mechanisms can function effectively.

This differentiated approach requires electoral management bodies to diagnose local ecologies before deploying resources, a capacity-intensive undertaking but one that vastly improves the efficiency of enforcement efforts. The alternative—uniform national standards applied to heterogeneous territories—produces either over-enforcement in low-risk environments or under-enforcement where it matters most.

7.5 Broader Implications and Future Research

Taken together, these findings advance a territorial theory of institutional performance in transitional settings. The success of inclusionary designs depends less on their formal architecture than on their fit with the uneven political geographies into which they are inserted. Autonomy is not legislated into existence through rules; it is constructed through the interaction of institutional design, organizational capacity, and state presence. Where these elements align, even fragile democracies can carve spaces of representation free from elite domination. Where they do not, formal insulation devolves into selective permeability, and the promise of political reparation remains unfulfilled.

The diagnostic framework developed here—operationalizing insulation as a testable restriction, using machine learning to recover configurations of influence, and translating patterns into actionable typologies—offers a portable template for evaluating inclusionary institutions in other fragmented states. The approach is particularly well-suited to post-conflict settings where the number of potential interactions far exceeds available theory, and where territorial heterogeneity demands flexible empirical tools. Future research should extend this framework to other reserved seat systems in Latin America (Peru's subnational indigenous constituencies) (Paredes and Došek 2020), sub-Saharan Africa (Uganda's special interest groups) (V. Wang and Yoon 2018), and South Asia (India's scheduled caste/tribe reservations, Pakistan's minority seats) (Krook 2009) to test whether the configurational patterns observed in Colombia generalize across institutional and regional contexts.

A second research agenda concerns the temporal dynamics of influence ecologies. The six-class typology identified here is a snapshot of municipal environments in 2022, but these ecologies are not static. *Mobilized Frontiers* may consolidate into *Institutionalized Enclaves* as state capacity scales and political machines adapt, or fragment into *Deprived Peripheries* if PDET infrastructure fails to materialize. *Expanding Community Councils* may prove resilient to co-optation over time, or they may face erosion as younger generations migrate and traditional authority structures weaken. Panel data tracking municipalities across multiple electoral cycles would allow researchers to model ecology transitions and identify the critical junctures at which local orders shift from autonomy to capture or vice versa.

Finally, the organizational dominance finding invites deeper inquiry into the internal structure of victim movements. Not all social movements are equally resistant to co-optation, and not all community councils possess equivalent gatekeeping capacity. What organizational characteristics—decision-making procedures, leadership selection mechanisms, resource autonomy, ties to NGOs or international donors—predict resilience to external influence? How do movements maintain horizontal coordination across municipalities without creating hierarchical vulnerabilities that machines can exploit? Qualitative case studies paired with network analysis of inter-organizational ties would complement the municipal-level diagnosis presented here, providing micro-level evidence on the mechanisms through which organizational density translates into political insulation.

The CITREP experiment reveals that designing inclusion in post-conflict settings is not a purely juridical exercise. It is a political and organizational challenge that requires matching formal rules to territorial realities, investing in the social infrastructures that sustain autonomy, and accepting that identical institutions will function differently across space. Understanding these differences is the first step toward building electoral systems that fulfill their democratic promise rather than reproducing the inequalities they were meant to remedy.

7.6 Applying the Framework Beyond Colombia

The diagnostic strategy developed here is designed for portability. While the Colombian case provides empirical grounding, the three-stage analytical sequence—residual-based anomaly detection, Random Forest feature importance with SHAP values, and Latent Class Analysis for typology construction—generalizes to any setting where identical institutional rules produce spatially variable outcomes. The framework does not require causal identification or controlled comparisons; it requires only measurable institutional outcomes, territorial covariates, and sufficient spatial variation to support classification algorithms. This makes it particularly well-suited to post-conflict and transitional settings where traditional causal inference faces insurmountable confounding.

Consider three cases where reserved seat systems exhibit similar patterns of uneven insulation. In **India's Scheduled Tribe (ST) reservations**, the diagnostic would operationalize insulation as the condition that ST reserved seat electoral outcomes are orthogonal to general constituency outcomes, conditional on district characteristics. Stage one constructs residuals from models predicting ST seat vote shares using general seat patterns and district-level controls (tribal population density, literacy, urbanization). Municipalities with large positive residuals—where ST outcomes diverge more than predicted—are labeled as exhibiting autonomous representation; large negative residuals indicate co-optation or alignment with dominant parties. Stage two applies Random Forest classification to these anomaly labels, inputting district-level covariates: caste association density, presence of tribal autonomous councils, state administrative capacity indices, and party system fragmentation. SHAP values identify which features most strongly predict insulation versus capture. If organizational variables dominate (e.g., density of registered tribal associations, strength of autonomous district councils), this confirms that formal reservations alone do not guarantee substantive autonomy—territorial organizational capacity conditions outcomes. Stage three uses LCA on the full covariate set to recover influence ecologies: districts might cluster into types such as "Autonomous Tribal Strongholds" (high organization, weak party penetration), "Co-opted Reservations" (state capacity high but tribal organizations fragmented), or "Representational Voids" (both state and organizational capacity low). The typology becomes an input for differentiated monitoring: autonomous districts require minimal oversight; co-opted districts need anti-capture interventions; voids require capacity-building before reservations can function.

The same logic extends to **Uganda's special interest group seats** (women, youth, persons with disabilities, workers, military). Define insulation as the condition that special interest seat outcomes are statistically independent of general constituency results for the same district, controlling for socioeconomic characteristics. Residuals identify anomalies. Random Forest on district covariates—civil society organization density by interest category, district administrative capacity, ruling party vote share, historical conflict exposure—produces SHAP-ranked feature importance. If organizational variables (e.g., density of women's cooperatives, youth associations, disability rights groups) emerge as dominant predictors, this indicates stratified autonomy: spe-

cial interest representation depends on whether beneficiary groups possess autonomous organizing capacity. LCA typologizes districts into influence ecologies that combine organizational strength, state presence, and partisan alignment. In **Peru's subnational indigenous quotas** (regional and municipal reserved seats, not national parliament), insulation is operationalized as indigenous candidate vote shares uncorrelated with non-indigenous candidate patterns in the same jurisdiction. Random Forest inputs include indigenous organization density (presence of rondas campesinas, comunidades nativas, federations), municipal fiscal capacity, party system institutionalization, and geographic remoteness. SHAP values test whether organizational infrastructure or structural conditions dominate. LCA produces ecological types distinguishing municipalities where indigenous movements exercise autonomous gatekeeping from those where mestizo parties co-opt indigenous candidates or where weak organizations leave seats vacant or captured by brokers.

What this analytical portability demonstrates is that stratified autonomy is not a Colombia-specific pathology but a general property of inclusionary institutions operating under territorial heterogeneity. The diagnostic framework makes this property measurable, typologizable, and actionable. It converts the normative question "do reserved seats work?" into the empirical question "where do they work, and what distinguishes success from failure?" By identifying the organizational and institutional configurations that pattern outcomes, the framework provides a basis for evidence-based intervention: either design districts to maximize within-district homogeneity on organizational capacity, or differentiate oversight mechanisms to address distinct pathways of influence across ecological types. The alternative—deploying uniform institutions with uniform monitoring under the assumption that formal rules alone guarantee substantive inclusion—perpetuates the territorial inequalities these institutions were designed to overcome.

8 Conclusion

This paper examined a question central to the design of post-conflict institutions: can electoral mechanisms created to include historically excluded populations remain insulated from the political machines they were meant to replace? Colombia's Circunscripciones Transitorias Especiales de Paz offered a natural experiment in institutional engineering, embedding a strong insulation claim within a politically heterogeneous landscape. The 2022 elections revealed that this claim held unevenly. In some municipalities, victim organizations mobilized autonomously and achieved representation consistent with the reform's normative intent. In others, traditional political networks penetrated the peace constituencies despite formal exclusion rules, transforming instruments of inclusion into extensions of partisan control. The central contribution of this study is to show that this variation is not random but systematic, arising from predictable configurations of state capacity, organizational density, and territorial political economies.

Substantively, the findings demonstrate that the permeability of institutional firewalls is a conditional property of local political ecologies, not a rule-based outcome. The latent class analysis identifies six recurring municipal environments in which autonomy proves resilient, fragile, or illusory. Where territorially anchored community institutions provide the organizational substrate for representation, insulation succeeds even under material scarcity. Where capable local states are embedded in hybrid orders that blend administrative efficiency with clientelistic control, insulation fails despite formal capacity. And where economic frontiers, displacement, and illicit rents create arenas of active contestation, outcomes remain endpoint-sensitive, contingent on the balance between victim movements and traditional brokers. This typology reframes the normative debate: the problem is not that politics reasserts itself in transitional institutions, but that the structural conditions for autonomous politics are unevenly distributed across territories.

Theoretically, the study advances a configurational account of institutional performance in fragmented states. The Random Forest analysis reveals that organizational variables—the breadth of traditional party coalitions and the density of victim movements—dominate the predictive hierarchy, explaining more variance than state capacity, conflict exposure, or poverty. This empirical regularity underscores a core insight: influence operates through organizational asymmetries, not through economic coercion or administrative absence. Victim organizations enter electoral competition with vastly different capacities for autonomous mobilization, and those differences reflect prior histories of state engagement, armed conflict experience, and community federalism that have unevenly distributed organizational capital across space. Inclusionary designs that presume a level playing field ignore this heterogeneity and risk producing symbolic inclusion rather than substantive representation. The implication is that legal insulation must be paired with sustained investment in the organizational infrastructure that allows marginalized populations to act collectively and resist cooptation.

Methodologically, the paper demonstrates how interpretable machine learning can be integrated with comparative institutional analysis to diagnose complex political phenomena. The three-stage empirical strategy—design-aligned benchmark, Random Forest classification with SHAP interpretation, and latent class analysis—bridges the rigor of causal inference with the flexibility of exploratory discovery. The diagnostic baseline operationalizes the insulation claim as a falsifiable restriction, allowing us to detect anomalous comovement between peace-seat and congressional outcomes. The Random Forest approximates the latent function mapping municipal characteristics to influence, recovering nonlinear interactions and threshold effects that parametric models would miss. And the latent class analysis translates high-dimensional predictive patterns into a small set of interpretable ecologies, grounding the typology in both theory and data. This combination of supervised and unsupervised learning shows that algorithmic methods can be made transparent, substantively meaningful, and compatible with the explanatory standards of political science. The approach is particularly well-suited to post-conflict settings where the number of potential interactions far ex-

ceeds available theory, and where territorial heterogeneity demands flexible empirical tools.

The findings travel beyond Colombia to other transitional democracies implementing inclusionary electoral designs in contexts of uneven state capacity and entrenched clientelism. Reserved seats, special districts, and quota systems are increasingly deployed to redress historical exclusion in societies marked by deep territorial inequality—from indigenous constituencies in Latin America to ethnic quotas in sub-Saharan Africa to gender-based reservations in South Asia. The Colombian case suggests that the effectiveness of these mechanisms depends less on their formal design than on the subnational ecologies into which they are embedded. Where community institutions, social movements, or ethnic federalism provide pre-existing organizational infrastructures, insulation can succeed. Where administrative capacity is captured by hybrid orders, or where victim organizations lack the resources to compete with professional networks, formal exclusion rules prove insufficient. The diagnostic framework developed here—operationalizing insulation as a testable restriction, using machine learning to recover configurations of influence, and translating patterns into useful typologies—offers a portable template for evaluating inclusionary institutions in other fragmented states.

For policymakers and electoral authorities, the analysis clarifies where and how interventions can strengthen institutional firewalls. Generic oversight mechanisms—monitoring campaign finance, auditing candidate eligibility, deploying observers—apply uniform scrutiny to heterogeneous environments and risk misallocating enforcement resources. The typology identifies differentiated strategies tailored to specific ecologies: protecting community-led intermediation in federated councils, auditing the state-machine interface in institutionalized enclaves, lowering participation costs for victim organizations in contested frontiers, and investing in basic state-building in disconnected peripheries. This segmentation makes the oversight problem tractable by matching tools to the mechanisms through which influence operates in particular settings. It also underscores that protecting autonomy is not solely a matter of enforcement; it requires building the organizational and administrative infrastructures that make autonomy structurally feasible.

The implication for institutional design is unambiguous. Post-conflict electoral reforms cannot assume that identical rules produce identical outcomes across space. Territorial heterogeneity produces stratified autonomy: uniform institutions deliver protection unequally, with some communities protected by endogenous organizational capacity, others captured by machines operating through legal administrative channels, and still others abandoned in representational voids where neither autonomous movements nor traditional parties possess sufficient resources to compete. Territory is not a neutral container for institutional experimentation; it is a structuring force that conditions whether formal insulation translates into substantive autonomy. Designing for heterogeneity requires either constructing territorially homogeneous districts through empirical clustering of political-organizational characteristics, or implementing differentiated oversight mechanisms calibrated to the specific pathways through which influence operates in distinct ecologies. The alternative—deploying uniform

rules and uniform monitoring across diverse contexts—transforms inclusionary institutions from instruments of democratic deepening into mechanisms that reproduce the very territorial inequalities they were designed to transcend.

What the evidence reveals is sobering but not fatalistic. Under the right organizational and institutional alignments, even fragile democracies can carve spaces of representation free from elite domination. The challenge is to diagnose where those alignments exist, to invest in the infrastructures that sustain them, and to resist the temptation to treat inclusion as something that can be legislated into existence. Democratization in post-conflict settings is not a moment of institutional invention but a process of political negotiation, shaped by the interaction of design, territory, and organization. The Colombian peace constituencies forced that negotiation into public view, leaving empirical traces that we can measure, interpret, and learn from. That is the contribution this paper seeks to make: not a definitive judgment on success or failure, but a rigorous diagnosis of the conditions under which autonomy becomes possible.

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A Baseline Regression

Purpose

This appendix documents the baseline regression evidence from our prior study. Its role in the present manuscript is strictly motivational: it establishes that, contrary to a strong insulation claim, CITREP and House tallies are not orthogonal at the municipal level. We do not treat these estimates as part of the identification strategy in this paper.

Unit of analysis and data

The estimating sample is structured at the municipality–combination level: each observation pairs one CITREP party with one House list within a municipality, yielding $N = 332{,}321$ combinations. Municipal covariates are merged at the municipal level and repeat across combinations by construction.

Specification

We first estimate linear OLS models relating CITREP votes to House votes, conditioning on three municipal fundamentals (performance, intergovernmental transfers, and population). In those OLS models, variables are standardized to ease coefficient comparison. Diagnostic checks in that prior study indicated violations of linearity, normality, homoskedasticity, and autocorrelation assumptions, motivating a log–log specification estimated on the same dataset.

Results: OLS

Table 4 presents results of the OLS specification.

Table 4: OLS models of CITREP votes (standardized variables; municipality–combination unit).

	Model 1	**Model 2**	**Model 3**	**Model 4**
Hoouse Votes	0.036***	0.035***	-0.022***	-0.021***
	(0.002)	(0.002)	(0.002)	(0.002)
MDM		-0.025***	-0.023***	-0.022***
		(0.002)	(0.002)	(0.002)
SGP			0.025***	0.022***
			(0.002)	(0.002)
Population				0.119***
				(0.003)
Intercept	-0.000	-0.000	-0.000	-0.767***
	(0.002)	(0.002)	(0.002)	(0.028)
R^2	0.001	0.002	0.024	0.026
Adjusted R^2	0.001	0.002	0.024	0.026
N	332,321	332,321	332,321	332,321

Nota: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Note: * (p<0.05); ** (p<0.01); *** (p<0.001). Coefficients reflect standardized variables (z-scores).

Results: log–log specification

For variance stabilization and interpretability, as shown in table 5, the prior study applied natural logs to all variables and re-estimated the models on the same municipality–combination sample. Coefficients can be read as elasticities.

Table 5: Adjusted log–log models of CITREP votes (municipality–combination unit).

	Model 5	Model 6	Model 7	Model 8
log(House Votes)	0.07***	0.07***	0.03***	0.02***
	(0.00)	(0.00)	(0.00)	(0.00)
log(MDM)		0.37***	0.03***	0.03***
		(0.01)	(0.01)	(0.01)
log(SGP)			0.54***	0.24***
			(0.00)	(0.01)
log(Population)				3.50***
				(0.07)
Intercept	2.48***	1.06***	-10.29***	-11.68***
	(0.00)	(0.03)	(0.08)	(0.09)
R^2	0.01	0.02	0.08	0.09
Adjusted R^2	0.01	0.02	0.08	0.09
N	332,321	332,321	332,321	332,321

Note: * ($p<0.05$); ** ($p<0.01$); *** ($p<0.001$).

Interpretation

Two facts from the prior study motivate the present paper. First, the elasticity of CITREP votes with respect to House votes is positive and precisely estimated in bivariate form and remains statistically different from zero after conditioning on municipal fundamentals. Second, as fundamentals are added, the elasticity attenuates and model fit rises from approximately 0.01 to 0.09, indicating that part of the raw comovement reflects municipal scale and resources. We do not ascribe causal meaning to these coefficients in this manuscript; they serve to falsify a strong-form insulation claim and frame the need for a design that characterizes heterogeneous influence mechanisms.

B Latent Class Analysis: Formal Specification and Estimation

This appendix provides a formal description of the Latent Class Analysis (LCA) employed to construct the six-ecology typology presented in Section 6. The LCA translates the high-dimensional predictive patterns recovered by the Random Forest classifier into a small number of interpretable influence ecologies. Unlike the supervised Random Forest model, which predicts influence labels, the LCA is an unsupervised mixture model that identifies latent subgroups of municipalities based on the joint distribution of observed municipal characteristics. The convergence between the two approaches—the Random Forest identifying predictive importance and the LCA revealing configurational structure—provides internal validation that the influence ecologies are empirically grounded rather than artifacts of model specification.

B.1 Model Specification

The LCA models observed municipal characteristics as arising from a finite mixture of latent classes, each representing a distinct influence ecology. The model assumes that, conditional on class membership, observed indicators are locally independent. This conditional independence assumption implies that all associations among observed variables are explained by the latent class structure; municipalities within the same class are homogeneous with respect to their observed profiles, while heterogeneity across municipalities arises from their membership in different latent classes.

Formally, let $\mathbf{X}_i = (X_{i1}, \ldots, X_{iJ})$ denote the vector of observed municipal indicators for municipality i, where J is the total number of indicators, and let $Z_i \in \{1, \ldots, K\}$ be an unobserved latent variable indicating class membership. The observed indicators include measures of electoral comovement (residual-based influence flags), state capacity (municipal performance index, fiscal balance, central government transfers), socioeconomic vulnerability (rurality, poverty indices, population scale), conflict exposure (forced displacement, terrorist acts, coca cultivation), and organizational structure (representation by community councils, ethnic organizations, and social movements).

The LCA specifies the joint distribution of the observed indicators as a finite mixture:

$$\Pr(\mathbf{X}_i) = \sum_{k=1}^{K} \Pr(Z_i = k) \prod_{j=1}^{J} \Pr(X_{ij} \mid Z_i = k).$$

The model assumes conditional (local) independence of the indicators given latent class membership, such that:

$$\Pr(\mathbf{X}_i \mid Z_i = k) = \prod_{j=1}^{J} \Pr(X_{ij} \mid Z_i = k).$$

This factorization states that, within each latent class k, the observed indicators are independent. All covariation among indicators is therefore captured by the latent variable Z_i, which partitions municipalities into discrete homogeneous subgroups.

Each class k is characterized by a vector of class-specific response probabilities $\boldsymbol{\theta}_k = \{\Pr(X_{ij} = x \mid Z_i = k)\}$ for all indicators j and all possible values x of each indicator. For continuous indicators, we discretize the distributions using empirically meaningful cutpoints (e.g., terciles or quartiles) to maintain interpretability and computational tractability. The mixing proportions $\pi_k = \Pr(Z_i = k)$ represent the prior probability that a municipality belongs to class k, and they satisfy the constraint $\sum_{k=1}^{K} \pi_k = 1$.

B.2 Parameter Estimation

Given the observed data $\{\mathbf{X}_i\}_{i=1}^{N}$ for $N = 167$ municipalities, the model parameters $\{\pi_k, \boldsymbol{\theta}_k\}$ are estimated by maximizing the log-likelihood function:

$$\ell(\boldsymbol{\pi}, \boldsymbol{\theta}) = \sum_{i=1}^{N} \log \left(\sum_{k=1}^{K} \pi_k \prod_{j=1}^{J} \Pr(X_{ij} \mid Z_i = k; \boldsymbol{\theta}_k) \right).$$

Direct maximization of this likelihood is computationally challenging because the latent class memberships $\{Z_i\}$ are unobserved, rendering the likelihood a nonlinear function of the parameters. We therefore employ the Expectation-Maximization (EM) algorithm, a standard iterative procedure for maximum likelihood estimation in latent variable models Collins2010.

The EM algorithm alternates between two steps:

E-step (Expectation): Given current parameter estimates $\boldsymbol{\pi}^{(t)}$ and $\boldsymbol{\theta}^{(t)}$ at iteration t, compute the posterior class membership probabilities for each municipality using Bayes' rule:

$$\Pr(Z_i = k \mid \mathbf{X}_i; \boldsymbol{\pi}^{(t)}, \boldsymbol{\theta}^{(t)}) = \frac{\pi_k^{(t)} \prod_{j=1}^{J} \Pr(X_{ij} \mid Z_i = k; \boldsymbol{\theta}_k^{(t)})}{\sum_{\ell=1}^{K} \pi_\ell^{(t)} \prod_{j=1}^{J} \Pr(X_{ij} \mid Z_i = \ell; \boldsymbol{\theta}_\ell^{(t)})}.$$

These posterior probabilities represent the expected class memberships conditional on the observed data and current parameter estimates. They are used as soft assignments in the subsequent maximization step.

M-step (Maximization): Update the parameter estimates by maximizing the expected complete-data log-likelihood, weighted by the posterior class membership probabilities computed in the E-step. The updated mixing proportions are:

$$\pi_k^{(t+1)} = \frac{1}{N} \sum_{i=1}^{N} \Pr(Z_i = k \mid \mathbf{X}_i; \boldsymbol{\pi}^{(t)}, \boldsymbol{\theta}^{(t)}),$$

and the updated class-specific response probabilities are:

$$\Pr(X_{ij} = x \mid Z_i = k; \boldsymbol{\theta}_k^{(t+1)}) = \frac{\sum_{i=1}^{N} \mathbb{1}(X_{ij} = x) \cdot \Pr(Z_i = k \mid \mathbf{X}_i; \boldsymbol{\pi}^{(t)}, \boldsymbol{\theta}^{(t)})}{\sum_{i=1}^{N} \Pr(Z_i = k \mid \mathbf{X}_i; \boldsymbol{\pi}^{(t)}, \boldsymbol{\theta}^{(t)})},$$

where $\mathbb{1}(X_{ij} = x)$ is an indicator function equal to 1 if municipality i exhibits value x on indicator j, and 0 otherwise.

The E-step and M-step are iterated until convergence, defined as the point at which the change in log-likelihood between successive iterations falls below a pre-specified tolerance threshold (typically 10^{-6}). Because the EM algorithm can converge to local maxima, we run the estimation procedure with 500 random starting values and retain the solution with the highest log-likelihood, following best practices in latent variable modeling Linzer2011.

B.3 Model Selection

A critical decision in LCA is the choice of the number of classes K, which determines the granularity of the typology. We evaluate models with $K = 2, 3, \ldots, 8$ classes using a combination of statistical fit criteria and substantive interpretability. The primary fit statistics are:

Bayesian Information Criterion (BIC):

$$\text{BIC} = -2\ell(\hat{\boldsymbol{\pi}}, \hat{\boldsymbol{\theta}}) + p \log(N),$$

where p is the number of free parameters in the model and $N = 167$ is the sample size. Lower BIC values indicate better fit, penalized for model complexity. The BIC is particularly well-suited to mixture models because it adjusts for overfitting by imposing a stronger penalty than the Akaike Information Criterion (AIC).

Sample-Size Adjusted BIC (aBIC):

$$\text{aBIC} = -2\ell(\hat{\boldsymbol{\pi}}, \hat{\boldsymbol{\theta}}) + p \log\left(\frac{N+2}{24}\right),$$

which applies a less stringent penalty than BIC and is recommended for smaller sample sizes Nylund-Gibson2018.

Bootstrap Likelihood Ratio Test (BLRT): A Monte Carlo simulation-based test that compares a K-class model to a $(K-1)$-class model. A significant BLRT p-value indicates that the more complex model fits the data significantly better than the simpler alternative.

Entropy:

$$E = 1 - \frac{\sum_{i=1}^{N} \sum_{k=1}^{K} \Pr(Z_i = k \mid \mathbf{X}_i) \log \Pr(Z_i = k \mid \mathbf{X}_i)}{N \log(K)},$$

which measures classification certainty. Entropy ranges from 0 to 1, with values closer to 1 indicating clearer separation between classes. High entropy (above 0.80) suggests that municipalities are confidently assigned to their respective classes with minimal ambiguity.

Based on these criteria, we select the six-class solution ($K = 6$) as the optimal model. Compared to the four-class solution, the six-class model exhibits lower BIC

and aBIC values, indicating improved fit without excessive complexity. The BLRT strongly rejects the four-class model in favor of the six-class specification ($p < 0.001$). The six-class model achieves an entropy of 0.89, confirming clear class separation, with average posterior probabilities across classes ranging from 0.82 to 0.94. By contrast, solutions with $K = 7$ or $K = 8$ classes introduce overfitting: the BIC begins to increase, several classes exhibit unstable membership (fewer than 10 municipalities), and substantive interpretation becomes ambiguous as theoretically distinct profiles collapse into statistically indistinguishable clusters. The six-class solution therefore provides the optimal balance between statistical parsimony and substantive richness.

B.4 Fit Diagnostics

This section reports the model selection diagnostics used to determine the number of latent classes in the Latent Class Analysis (LCA). We compare specifications ranging from one to ten classes using standard information criteria, with the goal of balancing statistical fit, parsimony, and substantive interpretability.

Figure 4 plots the Bayesian Information Criterion (BIC) and the Akaike Information Criterion (AIC) across alternative class solutions. Both criteria decrease monotonically as additional classes are introduced, reflecting improved in-sample fit. However, the rate of improvement differs substantially across specifications. The BIC exhibits a sharp decline up to six classes and flattens thereafter, indicating diminishing marginal gains in fit beyond this point. By contrast, the AIC continues to favor more complex models, a well-known tendency given its weaker penalty for model complexity.

The divergence between BIC and AIC beyond six classes reflects a common trade-off in finite mixture models. While additional classes continue to capture idiosyncratic variation, they do so at the cost of increased model complexity and reduced interpretability. In the present application, solutions with seven or more classes yield only marginal improvements in fit while producing small, unstable classes with ambiguous substantive profiles. These higher-order solutions also complicate interpretation by fragmenting otherwise coherent municipal configurations identified by the Random Forest analysis.

In contrast, the six-class solution provides a clear improvement over more parsimonious alternatives while preserving a substantively meaningful typology. This specification aligns closely with the configurational logic developed in the theoretical framework and yields high classification certainty, with entropy values close to 0.9 and average posterior probabilities ranging from 0.82 to 0.94 across classes. Importantly, the six-class structure captures the principal modes of territorial heterogeneity without overfitting noise or generating analytically redundant categories.

Taken together, the information criteria, classification diagnostics, and substantive coherence of the resulting classes support the selection of a six-class solution. This specification therefore serves as the basis for the influence ecologies discussed in Section 6.

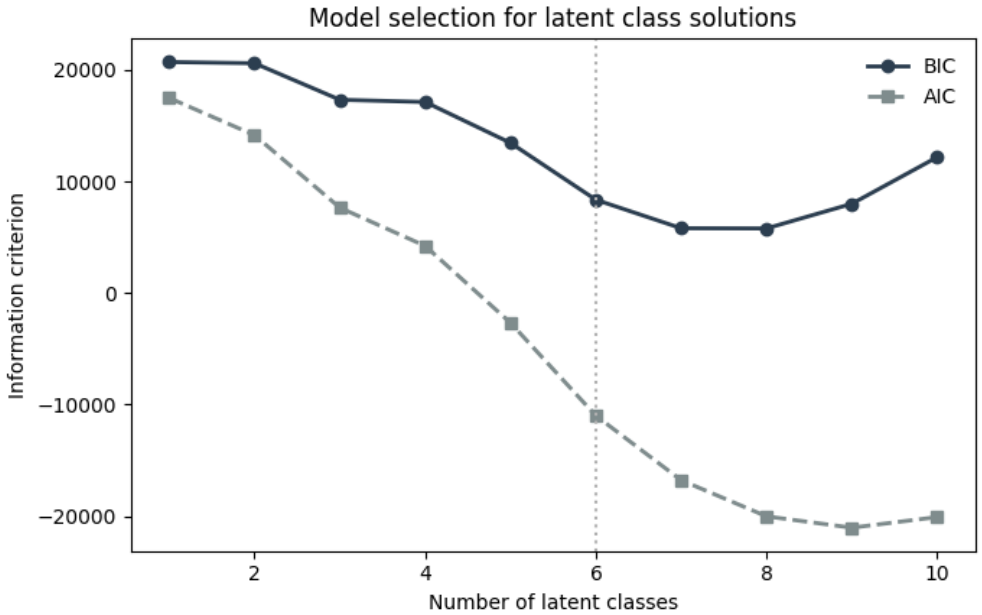

Figure 4: *Model selection for latent class solutions.* The figure reports the Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC) for models with one to ten latent classes. The vertical dashed line marks the six-class solution retained for the substantive analysis.

B.5 Class Assignment and Uncertainty

Once the optimal number of classes is determined, municipalities are assigned to classes based on maximum posterior probability:

$$\hat{Z}_i = \arg\max_{k \in \{1,\ldots,K\}} \Pr(Z_i = k \mid \mathbf{X}_i; \hat{\boldsymbol{\pi}}, \hat{\boldsymbol{\theta}}).$$

This modal assignment provides a hard classification that is used for descriptive summaries and spatial visualization. However, we retain the full posterior probability distribution $\{\Pr(Z_i = k \mid \mathbf{X}_i)\}_{k=1}^{K}$ for each municipality to account for classification uncertainty in subsequent analyses.

To ensure that the LCA measurement model is not distorted by external variables or outcomes, we employ a three-step approach Collins2010. In the first step, we estimate the unconditional LCA model to assign class membership based solely on the municipal indicators. In the second step, we compute posterior probabilities rather than treating class assignments as fixed, thereby accounting for classification uncertainty. In the third step, we relate class membership to external outcomes of interest—such as the Random Forest-predicted influence probabilities and observed CITREP vote shares—while correcting for measurement error in the class assignments using the posterior probabilities as weights. This procedure preserves the integrity of the latent class typology while allowing us to assess its external validity and substantive relevance.

B.6 Implementation

The LCA is implemented using the `poLCA` package (Linzer and Lewis 2011) in R, which provides robust estimation for polytomous latent class models via the EM algorithm. All continuous indicators are discretized into three to five categories using substantively meaningful cutpoints informed by the distribution of the data and theoretical expectations. Binary indicators (e.g., presence of community councils, ethnic representation) are included directly without transformation. Missing values, which are minimal in the administrative data sources employed, are handled through pairwise deletion during likelihood construction.

Model convergence is assessed by monitoring the log-likelihood trajectory across iterations. All models converge within 1,000 iterations for the majority of starting values, with the retained solution achieving convergence within 500 iterations. Sensitivity analyses confirm that the six-class structure is robust to alternative discretization schemes and to the exclusion of individual indicators with high missingness or low variance. The spatial distribution of latent classes is visualized using standard GIS software, overlaying class assignments onto municipality polygons to reveal geographic clustering and within-department heterogeneity.

In summary, the LCA provides a disciplined method for translating the Random Forest's high-dimensional predictive patterns into a compact, interpretable typology of influence ecologies. By modeling the joint distribution of municipal characteristics as a finite mixture, the LCA reveals the configurational structure underlying the heterogeneity in CITREP outcomes. The resulting six-class solution is statistically well-identified, substantively coherent, and aligned with the theoretical expectations developed in Section 3. This formal specification ensures that the influence ecologies presented in Section 6 rest on a rigorous empirical foundation rather than ad hoc clustering or descriptive intuition.

C Random Forest Feature Importance: Complete Rankings

Figure 5 reports mean absolute SHAP values for all municipal covariates included in the Random Forest classifier trained on the binary influence labels. SHAP values quantify the average contribution of each feature to the model's predictions across all municipalities, providing a comparative measure of explanatory importance. Higher values indicate that a variable consistently shifts predicted probabilities of influence, while lower values exert only marginal effects. Variables are ranked in descending order of importance and color-coded by tier: Tier 1 (dark) comprises core predictors with SHAP ≥ 0.024, accounting for the model's primary explanatory structure; Tier 2 (medium gray) includes contextual modifiers with SHAP values between 0.017 and 0.023 that refine classifications in specific environments; and the long tail (light gray) consists of minor predictors with SHAP < 0.017. The top ten predictors, which account

for 50.7% of total model importance, are presented and discussed in Section 6. The top eighteen variables collectively explain 66.6% of total variation. This appendix provides the complete ranking to document the full feature space and support reproducibility.

D Full Latent Class Results

The following tables report the full set of latent class profiles corresponding to the six-class solution. The table presents standardized mean values for all municipal covariates included in the latent class analysis, together with the relative size (*share*) of each class. This comprehensive presentation complements the core profiles discussed in the main text by documenting how the complete feature space aligns within each latent influence ecology.

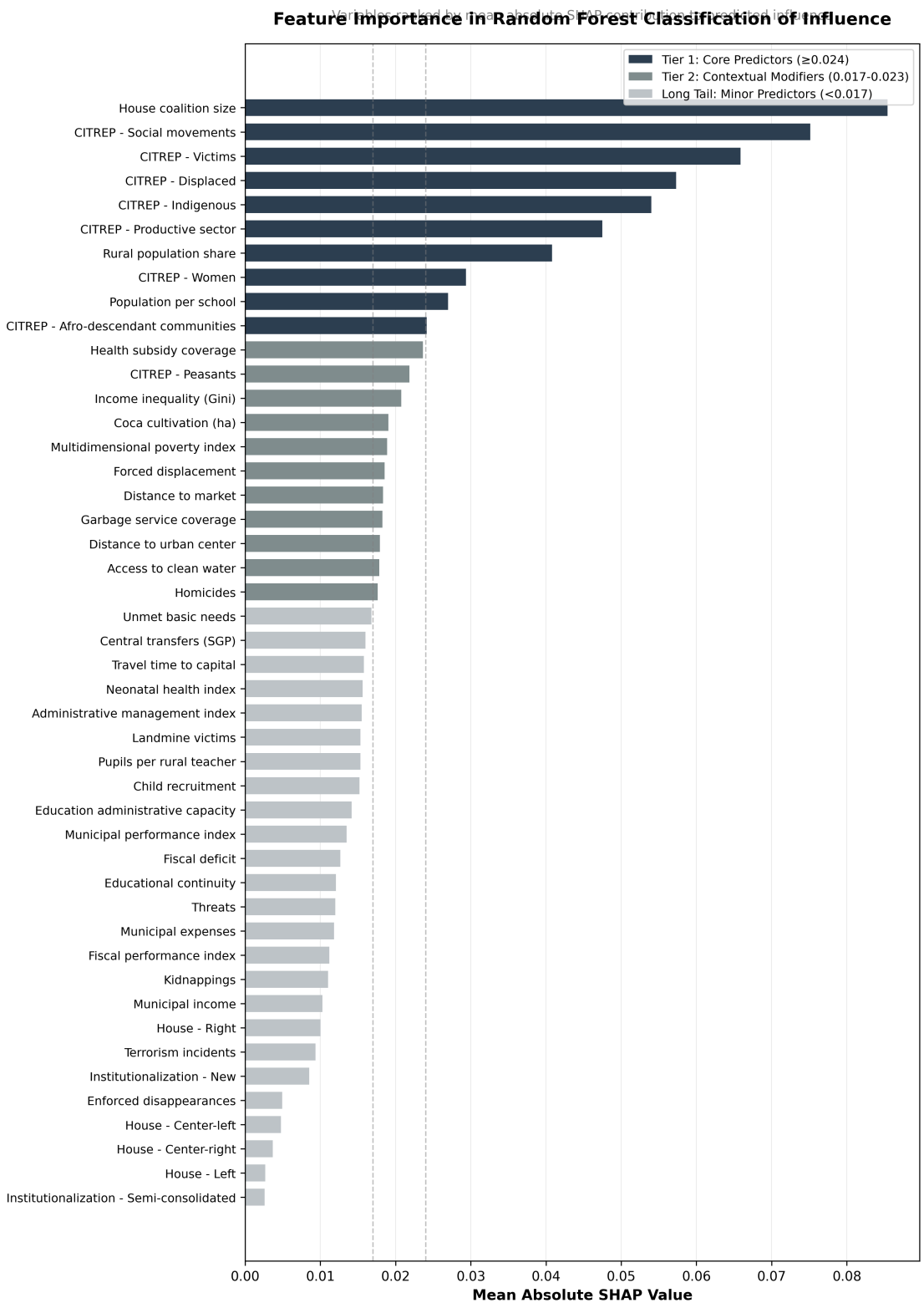

Figure 5: Complete Feature Importance Rankings from Random Forest Classification. Variables are color-coded by tier, with dashed vertical lines marking tier boundaries at SHAP values of 0.024 and 0.017. Organizational variables (coalition size and victim movement types) dominate the upper tier, while structural and conflict-related variables populate the middle and lower tiers.

Table 6: Latent class profiles: health, education, and administration

Latent class	Neonatal health	Admin. management	Pupils per rural teacher	Years of schooling
0	-0.001	-0.001	-0.001	-0.001
1	0.000	0.001	-0.001	0.002
2	0.001	0.000	0.001	0.003
3	0.003	0.000	0.000	0.001
4	0.003	0.003	0.006	0.004
5	-0.003	-0.002	-0.007	0.000

Table 7: Latent class profiles: public services

Latent class	Clean water	Sewerage	Garbage coverage	Health subsidy
0	0.001	0.000	0.002	0.001
1	0.002	0.001	0.002	0.006
2	-0.001	-0.001	-0.001	-0.001
3	0.011	0.003	0.007	0.011
4	0.006	0.003	0.005	0.012
5	-0.003	-0.004	-0.005	-0.006

Table 8: Latent class profiles: organizational representation

Latent class	Peasants	Community councils	Displaced	Indigenous
0	0.000	-0.002	0.002	0.008
1	-0.001	0.005	-0.006	0.004
2	0.000	-0.005	-0.002	0.000
3	0.000	-0.008	0.016	0.016
4	-0.007	-0.006	-0.005	0.001
5	-0.001	-0.005	-0.003	0.001

Table 9: Latent class profiles: political organization

Latent class	CITREP Women	Productive sector	Social movements	Coalition size
0	0.000	0.003	-0.011	-0.002
1	0.000	0.002	-0.007	0.009
2	0.004	0.004	-0.012	-0.003
3	-0.002	0.010	-0.008	0.001
4	0.013	-0.001	-0.008	0.005
5	0.002	0.003	-0.007	0.005

Table 10: Latent class profiles: conflict, territory, and class shares

Latent class	Forced displacement	Homicides	Rural population share	share
0	-0.001	0.000	0.013	0.287
1	0.001	0.001	0.009	0.132
2	-0.003	0.002	-0.013	0.246
3	0.009	0.004	0.023	0.042
4	0.006	0.006	0.016	0.096
5	-0.005	-0.002	-0.018	0.198